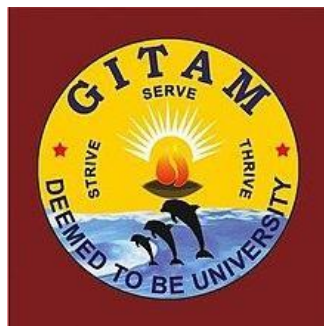


Application of Chemistry in Electronics (CHEM22411)

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TEXT BOOKS:

1. P.C. Jain and M. Jain, Engineering Chemistry, 15/e, ,Dhanapat Rai & Sons, 2014.
2. Electronic wastes by Hugo Marce;o veit and Andrea Moura , Bernardes, Springer, 2015, (ISBN978-319-15713-9)
3. O G Palanna, Engineering Chemistry, Tata McGraw Hill Education Private Limited, (2009)

References:

1. Sashi Chawla, A Textbook of Engineering Chemistry, Dhanapath Rai and sons, 2003.
2. B.S Murthy and P. Shankar, A Text Book of NanoScience and NanoTechnology, University Press 2013.
3. S.S. Dara, A Textbook of Engineering Chemistry, S. Chand & Co, 2010.
4. N. Krishna Murthy and Anuradha, A Text book of Engineering Chemistry, Murthy Publications 2014.



UNITS:

1. Water Technology
2. Energy Sources and Applications
3. Corrosion Engineering
4. Engineering Materials and Polymers
5. Nano and Smart Materials

Unit I: Advanced Electrochemical Energy Systems

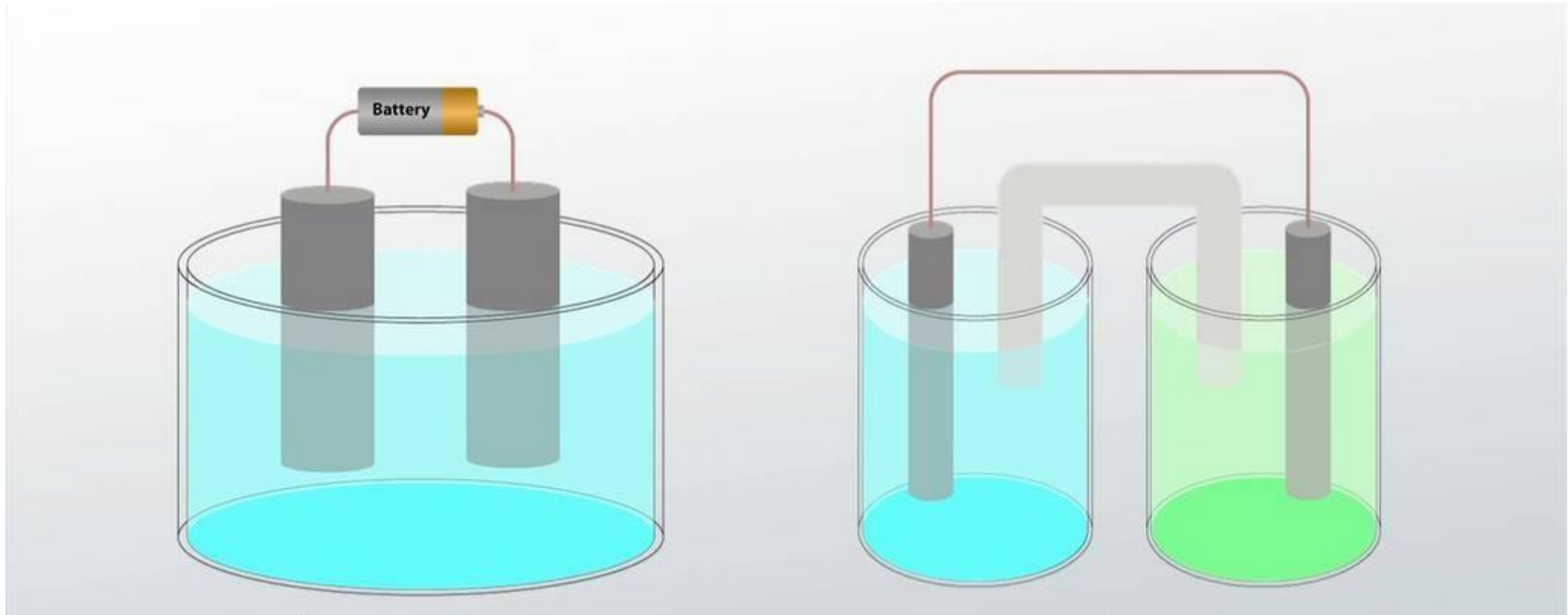
Advanced Electrochemical Energy Systems

9L

Electrodes – concepts, reference electrodes (Ag/AgCl electrode and glass electrode) electrochemical cell, cell potential calculations, numerical problems, Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples.

Primary cells –alkali metal sulphide batteries, Fuel cells, methanol fuel cells, Propane oxygen fuel cell – working of the cells.

Secondary cells – nickel-metal hydride - working of the battery including cell reactions, button cells.



Electrodes – concepts, reference electrodes (Ag/AgCl electrode and glass electrode)

Electrodes – Concepts

Electrodes are **conductive materials** that allow the flow of **electrons** into or out of an electrochemical system. They are essential components in **electrochemical cells** (galvanic and electrolytic cells).

Types of Electrodes:

1.Working Electrode: Where the main redox reaction occurs (e.g., platinum, carbon).

2.Counter (Auxiliary) Electrode: Completes the circuit and allows current to flow.

3.Reference Electrode: Maintains a **constant potential** against which the working electrode's potential is measured.

Reference Electrodes

Reference electrodes provide a **stable and known potential**, making them essential for accurate electrochemical measurements.

Characteristics:

- Constant potential
- Reproducible
- Non-polarizable
- Chemically inert to the system

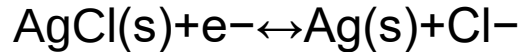
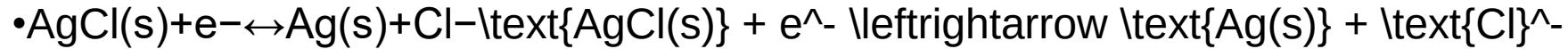
Two important types of reference electrodes are:

1. Ag/AgCl Electrode (Silver/Silver Chloride Electrode)

Construction:

- A **silver wire** coated with **silver chloride (AgCl)** is immersed in a **KCl solution** saturated with **AgCl**.

- Reaction at electrode:



Features:

- Simple to construct
- Stable potential (~ +0.197 V vs. SHE in 3.5 M KCl)
- Commonly used in aqueous system

Advantages:

- Inexpensive and nontoxic
- Low maintenance
- Good for biological and environmental measurements

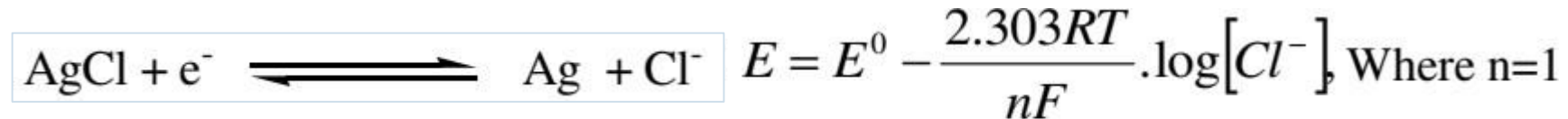
Silver- Silver Chloride electrode

- Silver-Silver chloride is also a metal-metal salt ion electrode.
- Silver and its sparingly soluble salt silver chlorides are in contact with a solution of chloride solution ions.
- Generally a silver wire is coated with AgCl and dipped in a solution of KCl .

Cell representation is as follows

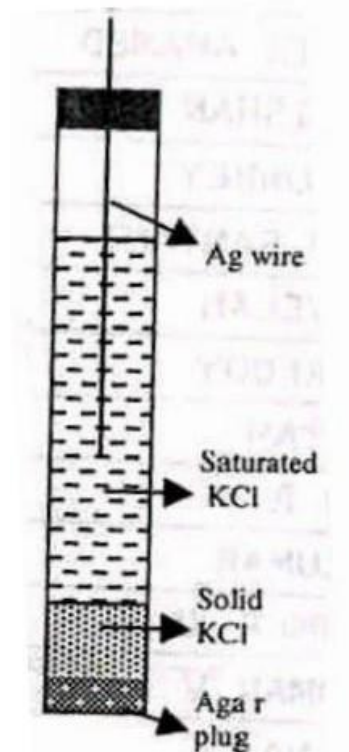


Net half cell reaction is



$$E = E^0 - \frac{2.303RT}{F} \log[\text{Cl}^-]$$

$$E = E^0 - 0.0591 \log[\text{Cl}^-] \text{ at } 298\text{K}$$



Therefore electrode potential of calomel electrode is depending upon the concentration of KCl.

- ✓ electrode is reversible with chloride ions.**
- ✓ potential of the calomel electrode depends on the concentration of the KCl.**

For 1N solution, the electrode potential is 0.223V and for saturated solution is 0.199V at 298K

Applications:

- 1. Used as secondary reference electrode in ion selective electrode.**
- 2. In determining the distribution of potential on the ship hull and pipe lines.**

2. Glass Electrode (pH Electrode)

Construction:

- A thin-walled **glass membrane** filled with an internal buffer solution (typically HCl), with an internal reference electrode (Ag/AgCl).
- Measures potential developed across the glass membrane due to **H⁺ ion activity**.

Working Principle:

- The glass membrane develops a potential that depends on the **H⁺ concentration** (pH) of the solution.
- The potential is measured relative to a reference electrode.

Key Equation:

$$E = E^\circ + 0.0591 \log[H^+] = E^\circ - 0.0591 \times \text{pH}$$
$$E = E^\circ + 0.0591 \log[H^+] = E^\circ - 0.0591 \times \text{pH}$$

Applications:

- Common in **pH meters** for laboratories, industries, and environmental monitoring.

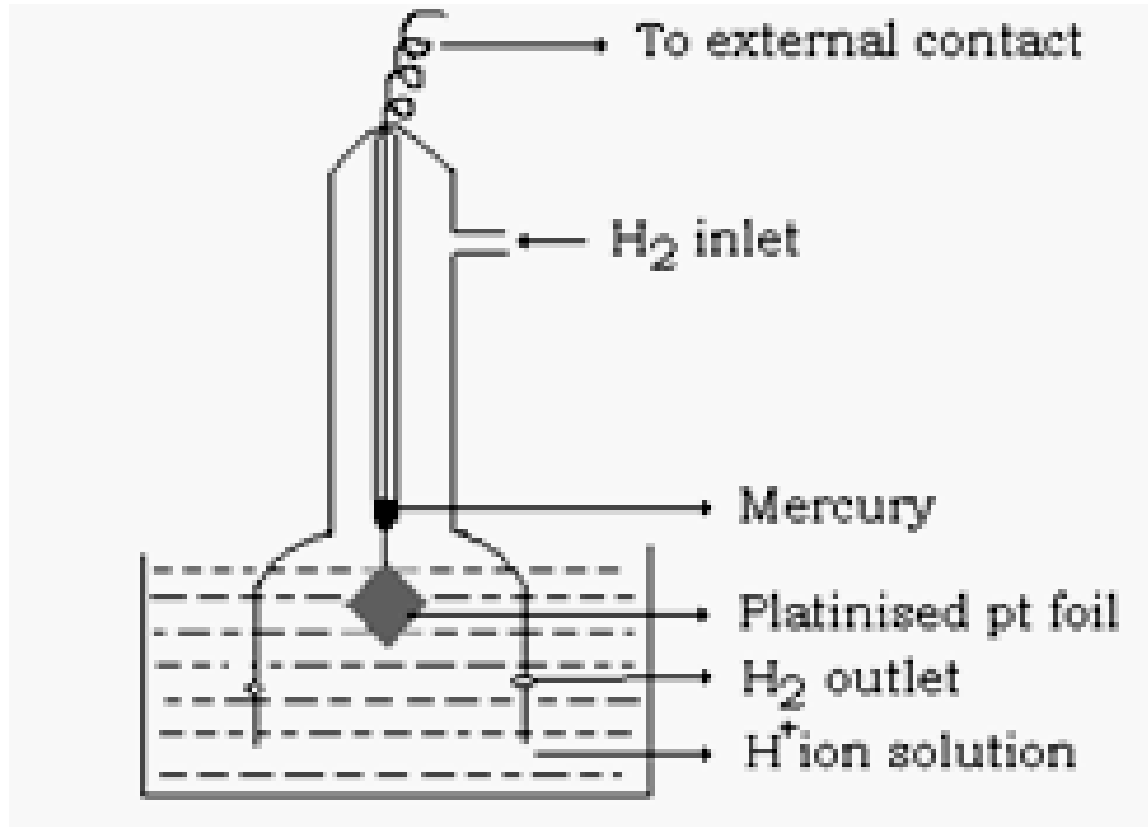
Advantages:

- High selectivity for H⁺
- Rapid response and wide pH range

Primary reference electrode:

- ✓ Standard Hydrogen Electrode (SHE) is primary reference electrode which is used for cell potential measurements.
- ✓ Its potential is taken as zero at all temperatures.

SHE Construction & working:



Electrode representation: $Pt/H_2 (g, 1 \text{ atm})/H^+ (1M)$

Reactions:

as anode: $H_2 (g, 1 \text{ atm}) \rightarrow 2H^+ (1M) + 2e^-$ ($E^0 = 0.00V$)

as cathode: $2H^+ (1M) + 2e^- \rightarrow H_2 (g, 1 \text{ atm})$ ($E^0 = 0.00V$)

Oxidation:

The process in which an atom or an ion loses electrons.

Reduction:

The process in which an atom or an ion gains electrons.

Redox reaction:

The chemical reactions in which oxidation and reduction occur simultaneously are called redox reactions.

Oxidising agent:

A species which oxidises others but gets reduced itself by gaining electrons is called an oxidizing agent.

Reducing agent:

A species which reduces others but gets oxidised itself by losing electrons is called a reducing agent

Electrode:

An electrode is a current conductor that passes an electrical current from one medium to another.

metals → copper, silver, lead or zinc, non-metallic substance → such as graphite.

Anode:

It is the electrode at which oxidation takes place.

Cathode:

It is the electrode at which reduction takes place.

Electrolyte:

It is the substance which conducts electricity either in the molten state or solution state.

Electrolysis:

It is phenomenon in which an electrolyte undergoes decomposition upon passing electricity and products are discharged at the electrodes.



CONVENTIONAL ENERGY SOURCES

- generation of electrical energy from conventional energy sources like electrochemical systems i.e. battery or fuel cells.
- these electrochemical systems work on electrochemistry
- which deals with two important phenomena as follows

IONICS:

- effect of electrical voltages and currents on chemical reactions
- governed by *Faraday's laws*.
- Ex: electrolytic cell (Electroplating process and mobile battery charge)

ELECTRODICS:

- voltages and currents are produced by chemical changes
- Governed by *Nernst's equation*.
- Ex. galvanic cell (mobile battery discharge)

Electrochemistry: A branch of chemistry which deals with inter-conversions between chemical energy $\leftrightarrow$ electrical energy.

An electrochemical cell – “a device” -- convert chemical energy $\leftrightarrow$ electrical energy.

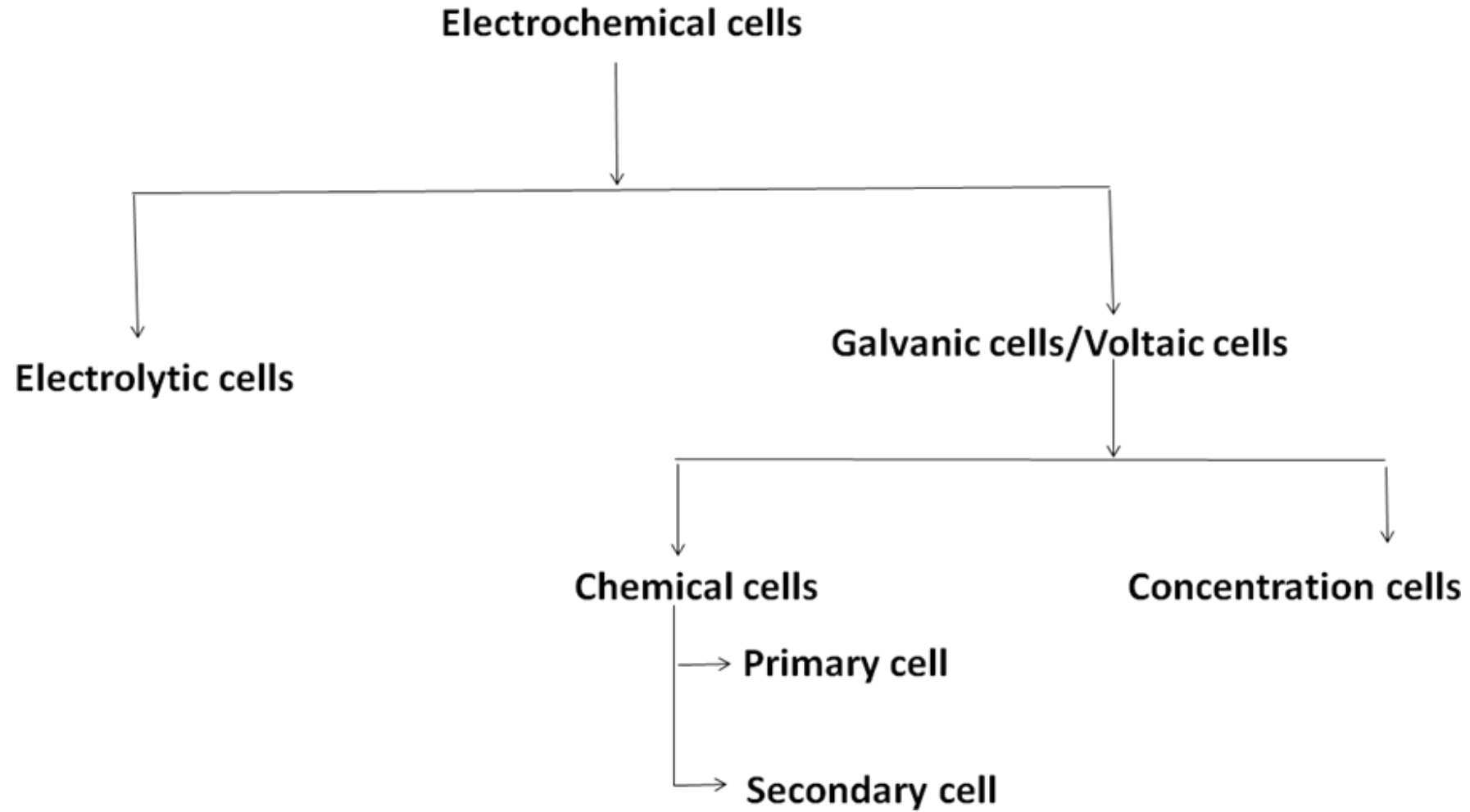
Types of electrochemical cells: Two

1) Galvanic or Voltaic cells: convert chemical energy into electrical energy.

Ex. Daniel cell, Dry cell, etc

2) Electrolytic cells: convert electrical energy into chemical energy.

Example: Electrolysis of molten NaCl



Chemical cells: A chemical cell converts chemical energy into electrical energy.

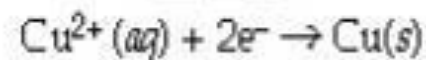
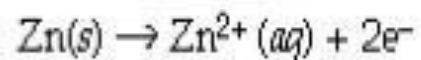
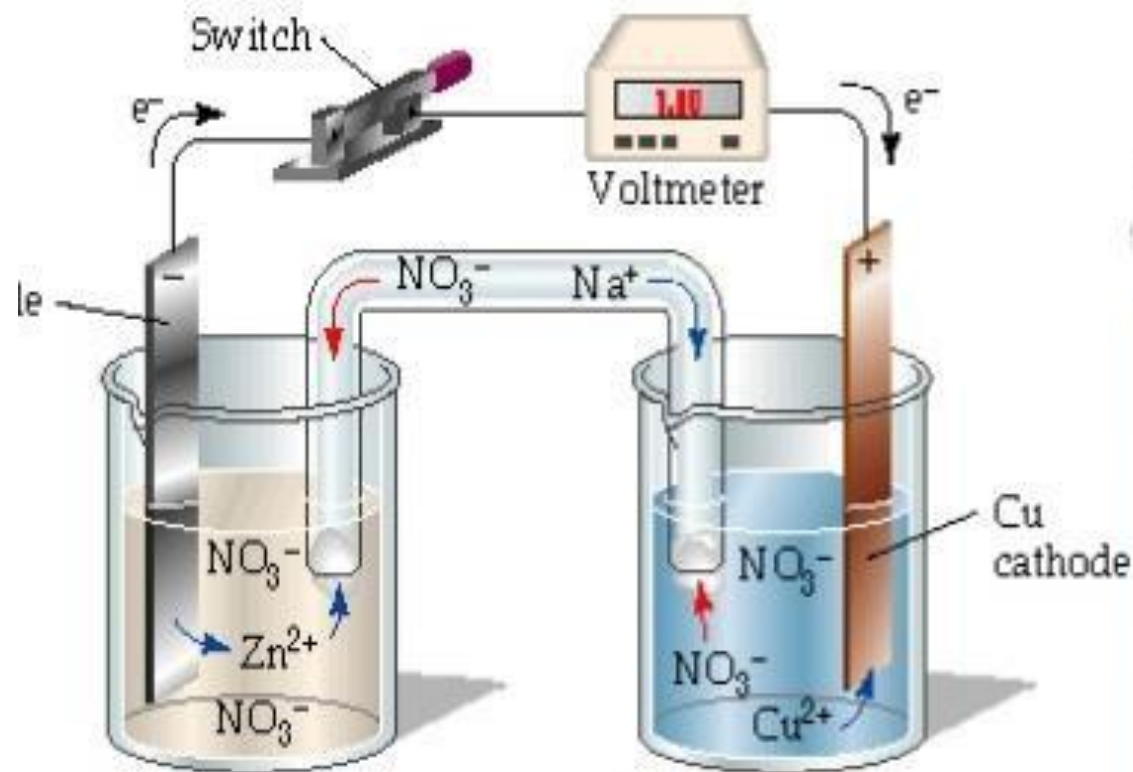
i) **Primary cells**: It is an electric cell that produces current by an irreversible chemical reaction.

ii) **Secondary cells**: It is an electric cell that can be recharged and can therefore be used to store electrical energy in the form of chemical energy.

Concentration cells: A concentration cell is an electrolytic cell that is comprised of two half-cells with the same electrodes, but differing in concentrations.



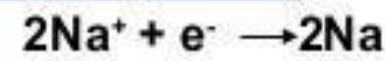
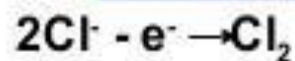
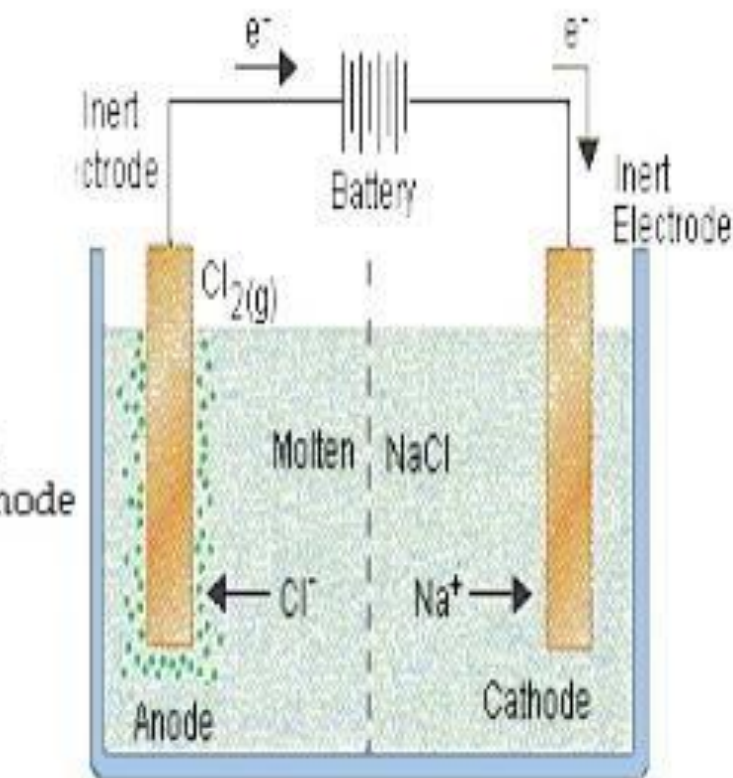
Differences between Electrolytic cell and Galvanic cell



Movement of cations

Movement of anions

Voltaic Cell

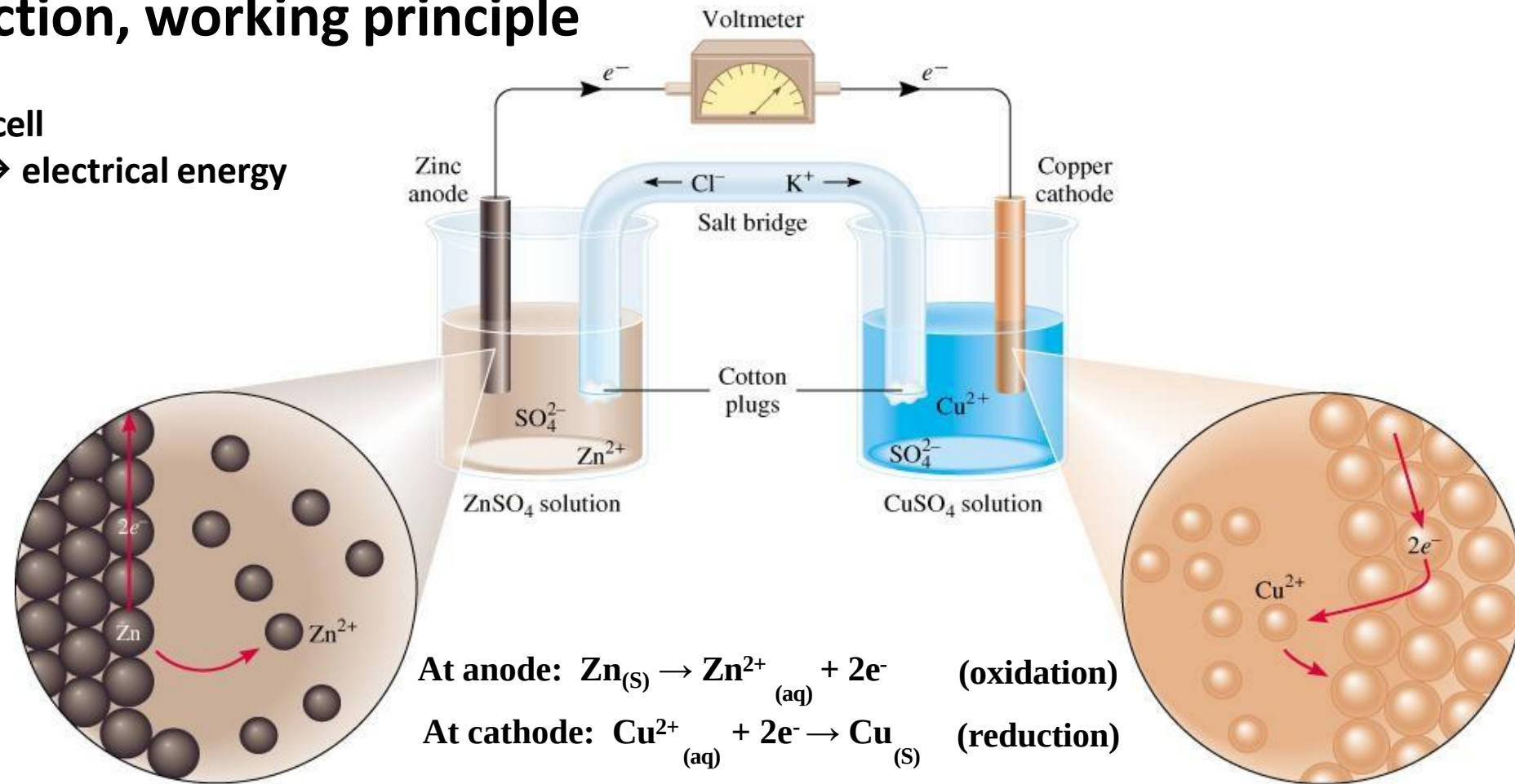


Electrolytic Cell

	Galvanic cell/Voltaic cell	Electrolytic cell
Definition	Spontaneous redox process that convert chemical energy into electrical energy	Non spontaneous redox reactions which converts electricity into chemical energy
Technique	Electricity is produced with the aid of chemical reaction	Electric current is used for development of a chemical reaction, utilizing an external source along the way
Design	The two half cells are setup in different containers and are connected through salt bridge (or) porous partition	The electrodes are in the same container in solution/molten state
Electrode Polarity	The anode is the negative and the cathode is positive	The anode is positive and cathode is negative electrode
Chemical reactions	The oxidation reaction takes place at the anode (Negative electrode) and the reduction is takes place at cathode (Positive electrode)	The oxidation reaction takes place at the anode (positive electrode) and the reduction is takes place at cathode (Negative electrode)
Applications	Batteries and Accumulators	Electroplating, Metal purification methods

Daniel cell- construction, working principle

- best example of a galvanic cell
- converts chemical energy → electrical energy
- Ex. Lead acid battery

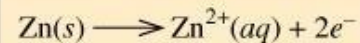


At anode: $\text{Zn}_{(s)} \rightarrow \text{Zn}^{2+}_{(aq)} + 2e^-$ (oxidation)

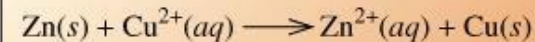
At cathode: $\text{Cu}^{2+}_{(aq)} + 2e^- \rightarrow \text{Cu}_{(s)}$ (reduction)

Net cell reaction: $\text{Zn}_{(s)} + \text{Cu}^{2+}_{(aq)} \rightarrow \text{Zn}^{2+}_{(aq)} + \text{Cu}_{(s)}$

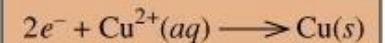
- Between the two half cells
- galvanometer shows a deflection
- bulb it starts glowing
- Voltammeter it measures the voltage.



Net reaction



Cu^{2+} is reduced to Cu at cathode.



Electromotive force (emf of a cell):

“The difference in potential that drives the e⁻s from one half cell → another half cell in a galvanic cell is called as electromotive force”.

It is denoted by E_{cell} .

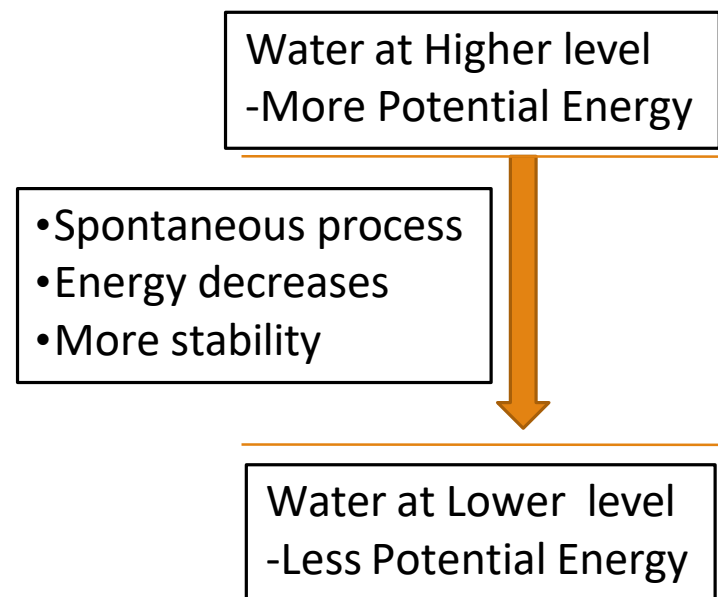
Salt bridge:

U-shaped glass tube containing a suitable electrolyte (Like KCl, KNO₃, K₂SO₄, etc) gelled by adding agar-agar.

The gel holds the electrolyte in the tube.

The role of salt bridge is,

- internal contact between two electrodes of a galvanic cell
- minimises polarisation of electrodes, by neutralising the charge of solutions
- removes liquid-liquid junction potential



i) Electrode Potential or Single electrode potential (E):

“It is the potential developed or potential difference that exists when a metal electrode or a glass is in contact or in equilibrium with the solution of its own ions”.

OR

“It is the measure of tendency of an electrode to lose or gain electrons when it is in contact with solution of its own ions”

ii) Oxidation potential:

“It is the potential developed or the potential difference exists due to oxidation reaction when a metal electrode or a gas is in contact or in equilibrium with the solution of its own ions”.

OR

“It is the measure of tendency of an electrode to lose electrons and undergo oxidation, when it is in contact with solution of its own ions”

iii) Reduction potential:

“It is the potential developed or potential difference exists due to reduction reaction when a metal electrode or a gas is in contact or in equilibrium with the solution of its own ions”.

OR

“It is a measure of tendency of an electrode to gain electrons and undergo reduction, when it is in contact with solution of its own ions”

iv) Standard electrode potential:

“It is the potential developed or potential difference exists when a metal electrode in contact or in equilibrium with 1M solution of its own ions at 298 K”

OR

“It is the potential developed when an inert electrode is in contact with a gas at 1 atmospheric pressure and with 1M solution of its own ions at 298 K”.

v) Standard Reduction Potential (SRP):

“It is a measure of tendency of a metallic electrode to gain electrons and undergo reduction, when it is in contact with solution of its own ions of 1 M concentration at 298K”.

vi) Standard Oxidation Potential(SOP):

“It is a measure of tendency of a metallic electrode to lose electrons and undergo oxidation, when it is in contact with solution of its own ions of 1 M concentration at 298K”.

Origin of Electrode Potential- Nernst Theory of electrode potential

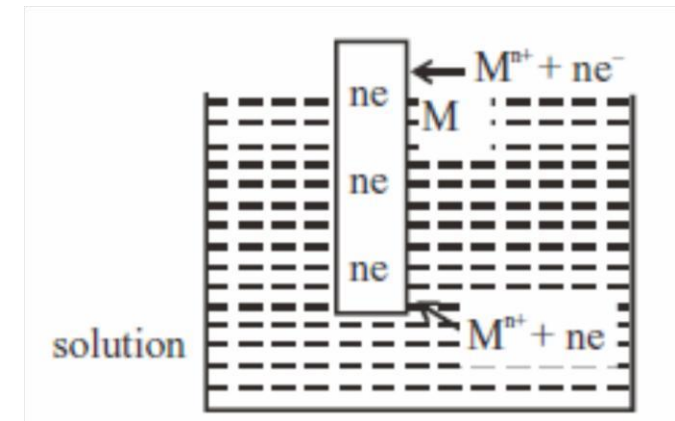
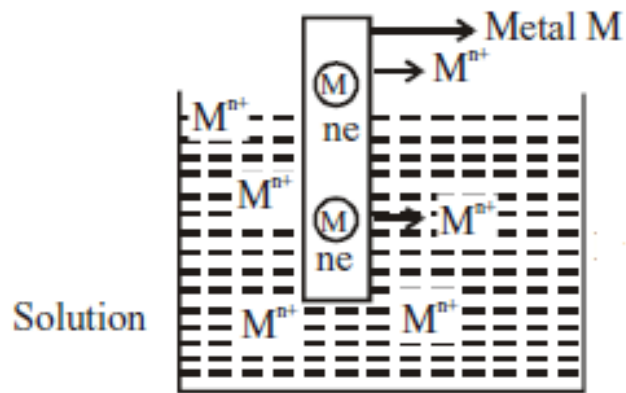
When a metal (M) is placed in a solution of its own salt (M^{n+}) one of the two processes are possible

Electrolytic solution pressure (ESP)

(i) Metal atoms go into solution in the form of ions.
 $M \rightarrow M^{n+} + ne^-$ (Oxidation)

-Osmotic pressure (OP)

(ii) Metal ions from solution may deposit on the metal
 $M^{n+} + ne^- \rightarrow M$ (reduction)



The tendency of metal to undergo oxidation or reduction depends on the magnitude of its electrolytic solution pressure and osmotic pressure.

Case (i) If $ESP > OP$, then it undergoes Oxidation.

Case(ii) If $ESP < OP$, then it undergoes reduction and

Case(iii) If $ESP = \text{Osmotic pressure}$, then it is said to be equilibrium

Origin of Electrode Potential- Nernst Theory of electrode potential

Example : Consider Zn metal Placed in the solution of ZnSO₄ solution

For Zn , $\longrightarrow$ At. No.= 30 , No. e⁻ = 30

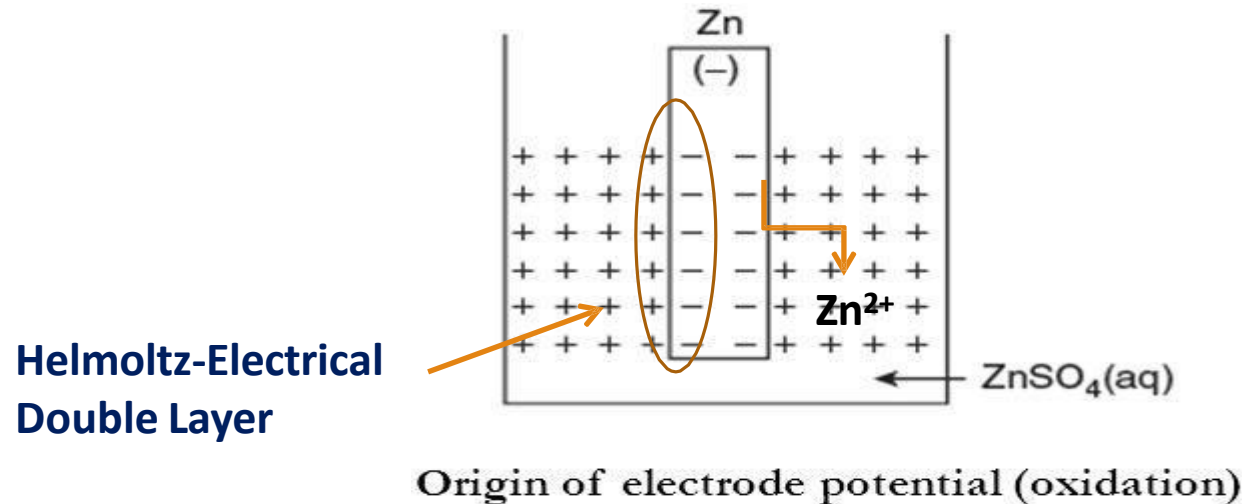
Electron configuration : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$ Less stable

For Zn²⁺ $\longrightarrow$ At. No.= 30, No. e⁻ = 28

Electron configuration : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{10}$ More stable

Which one is more stable Zn or Zn²⁺ ?

Thus Zn prefers to undergo oxidation or ESP > OP for Zn



Origin of Electrode Potential- Nernst Theory of electrode potential

Example : Consider Cu metal Placed in the solution of CuSO_4 solution

For Cu , $\longrightarrow$ At. No.= 29, no. e- = 29

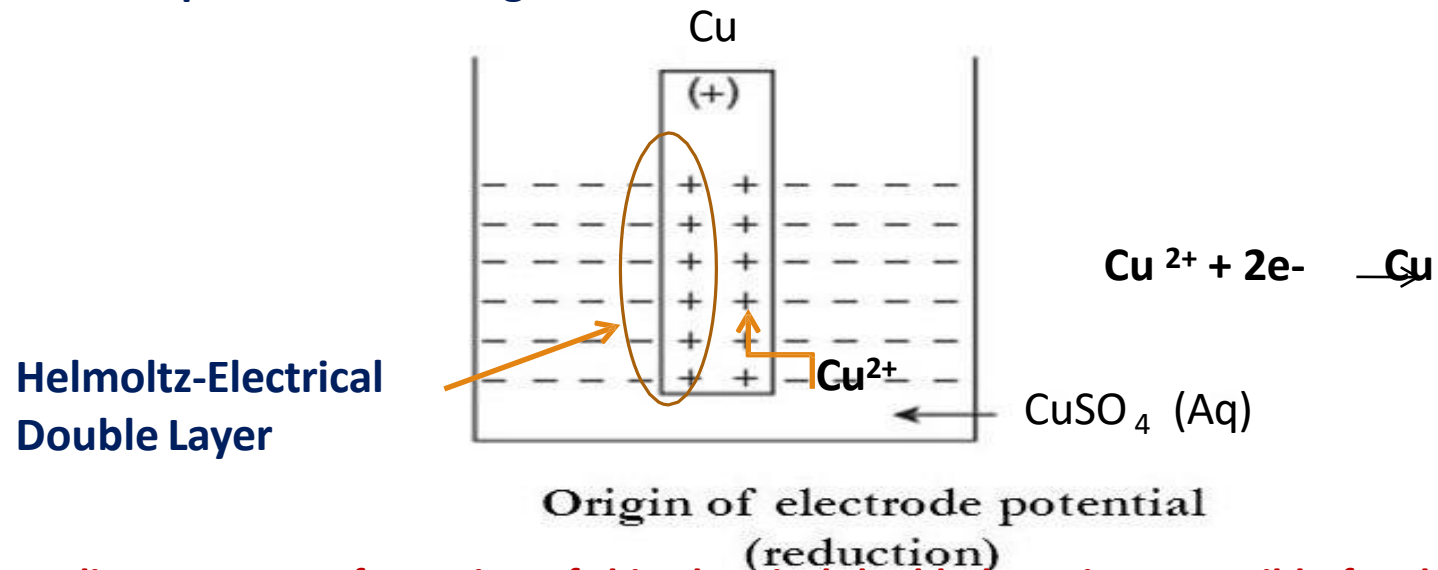
Electron configuration : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ **More stable**

For Cu^{2+} $\longrightarrow$ At. No.= 29, no. e- = 27

Electron configuration : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^9$ **Less stable**

Which one more stable Cu or Cu^{2+} ?

Thus Cu prefers to undergo Reduction or $\text{ESP} < \text{OP}$ for Cu



According to Nernst, formation of this electrical double layer is responsible for the origin of electrode potential.

Electromotive force (EMF):

It is defined as the potential difference between the two electrodes of a galvanic cell which causes the flow of current from an electrode with higher reduction potential to the electrode with lower reduction potential.

It is denoted as E_{cell} .

$$E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}.$$

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}.$$

Standard emf of a cell is defined as the emf of a galvanic cell when the reactants and products of the cell reaction are at unit concentration or unit activity at **298 K** and at **1 atmospheric pressure**.

The emf of a cell (E_{cell}) represents the driving force of a cell reaction.

As represented by the thermodynamic relation,

$$\Delta G = -nfE_{\text{cell}}$$

ΔG = free energy change accompanying the cell reaction

n = no. of electron transfer during the cell reaction

F = Faraday

From the above equation,

(i) For a galvanic cell, if E_{cell} is positive, then ΔG is negative and the cell reactions are spontaneous.

(ii) For a galvanic cell, if E_{cell} is negative, then ΔG is positive and the cell reactions are Non-spontaneous.

Nernst equation for single electrode potential

- During the construction of an electrode, **it is always not possible** to restrict concentration of the electrolyte to unit molar.
- In addition, the composition, temperature of the reaction also decides the electrode potential.
- Hence it is desired to have a quantitative equation relating single electrode potential (E), free energy change (ΔG) and concentration i.e $[M^{n+}]$.

In 1889, Nernst derived a quantitative relationship between these terms as follows.

According to thermodynamics, decrease in free energy ($-\Delta G$) represents the maximum amount of work available from a reversible chemical process (half cell).

$$-\Delta G = W_{\max} \quad (1)$$

The maximum electrical work done by an electrode depends on two factors.

(a) Number of coulombs of charge flowing across the metal electrode-electrolyte interface
&

(b) Energy available per every coulomb

Therefore, $W_{\max} = (\text{No. coulombs of charge flowing}) \times (\text{energy available per coulomb})$

$$W_{\max} = nF \times E \quad (2)$$

Nernst equation for single electrode potential

Comparing equations (1) and (2),

$$-\Delta G = nFE$$

$$\Delta G = -nFE \quad (3)$$

Similarly, $\Delta G^0 = -nFE^0 \quad (4)$

Where ΔG^0 is standard free energy change and E^0 is standard electrode potential.

For the reaction,



According to law of mass action, $K_c = \frac{[M]}{[M^{n+}]}$

{because ,the equilibrium constant $K_c = \frac{k_f}{k_b}$ and $k_f \propto [M^{n+}]$ and $k_b \propto [M]$ }

but, $[M] = 1$, because concentration of electrode is very large compared to metal ion.

$$\text{Therefore, } K_c = \frac{1}{[M^{n+}]} \quad (5)$$

The equilibrium constant, K_c is related to change in free energy by the Vant-hoff's equation.

$$\Delta G = \Delta G^0 + RT \ln K_c \quad (6)$$

Substituting eq. (3) and (4) in eq. (6) we have,

$$-nFE = -nFE^0 + RT \ln \frac{1}{[M^{n+}]}$$

Divide the above equation by nF and multiplying it by - throughout.

Nernst equation for single electrode potential

$$E = E^0 - \frac{RT}{nF} \ln \frac{1}{[M^{n+}]}$$

$$E = E^0 + \frac{RT}{nF} \ln [M^{n+}] \quad (7)$$

$$E = E^0 + \frac{0.0591}{n} \log [M^{n+}]$$

-Nernst equation for single electrode potential

Factors affecting electrode potential

From Nernst equation, it is clear that the electrode potential depends on,

- Metal electrode (E^0 value)- ex: $E^0_{Zn/Zn^{2+}} = -0.76 \text{ V}$, $E^0_{Cu^{2+}/Cu} = 0.34 \text{ V}$ etc
- Temperature- As temperature increases, the electrode potential also increases.
- Concentration of solution (M^{n+}):- As the concentration increases, electrode potential also increases.

Nernst equation for Cell potential

We know that, $E_{\text{cell}} = E_c - E_a$

Now substituting for E_c & E_a from equation (7),

$$E_{\text{cell}} = \left(E_c^0 + \frac{RT}{nF} \ln[\text{Species at the cathode}] \right) - \left(E_a^0 + \frac{RT}{nF} \ln[\text{Species at the anode}] \right)$$

$$E_{\text{cell}} = E_{\text{cell}}^0 + \frac{RT}{nF} \left(\ln[\text{Species at the cathode}] - \ln[\text{Species at the anode}] \right)$$

$$E_{\text{cell}} = E_{\text{cell}}^0 + \frac{0.059}{n} \log \frac{[\text{Species at the cathode}]}{[\text{Species at the anode}]}$$

(8) -Nernst equation for cell potential

Nernst equation for Daniel cell

$$E_{\text{cell}} = E^0 + \frac{0.059}{n} \log \frac{[\text{Cu}^{2+}]}{[\text{Zn}^{2+}]}$$

Applications of Nernst Equation:

- One of the major application of Nernst equation is in determining ion concentration
- It is also used to calculate the potential of an ion of charge "Z" across a membrane.
- It is also used in solubility products and potentiometric titrations.
- It is also used in pH measurements.

Problems on Nernst equation:

- (i) A zinc rod is dipped in 0.1M solution of ZnSO_4 . The salt is 95% dissociated at 298K. Calculate the electrode potential.

Solution: Concentration of Zn^{2+} ions = $0.1 \times 95/100$ M
 $[\text{Zn}^{2+}] = 0.095$ M

$$\begin{aligned}E_{\text{Zn}/\text{Zn}^{2+}} &= E_{\text{Zn}/\text{Zn}^{2+}}^0 + \frac{RT}{nF} \ln [\text{Zn}^{2+}] \\E_{\text{Zn}/\text{Zn}^{2+}} &= -0.76 + 0.0591/2 \log [0.095] \\E_{\text{Zn}/\text{Zn}^{2+}} &= -0.76 + -0.0302 \\E_{\text{Zn}/\text{Zn}^{2+}} &= -0.7902 \text{ V}\end{aligned}$$

- (ii) Calculate the electrode potential of copper wire dipped in 0.1 M CuSO_4 solution at 25°C. The standard reduction potential of Cu is 0.34 V.

Solution: Concentration of Cu^{2+} ions = 0.1 M
 $[\text{Cu}^{2+}] = 0.1$ M

$$\begin{aligned}E_{\text{Cu}^{2+}/\text{Cu}} &= E_{\text{Cu}^{2+}/\text{Cu}}^0 + \frac{RT}{nF} \ln [\text{Cu}^{2+}] \\E_{\text{Cu}^{2+}/\text{Cu}} &= 0.34 + 0.0591/2 \log [0.1] \\E_{\text{Cu}^{2+}/\text{Cu}} &= 0.34 + (-0.02955) \\E_{\text{Cu}^{2+}/\text{Cu}} &= 0.31045 \text{ V}\end{aligned}$$

- (iii) At what pH of HCl solution will H₂ gas electrode show an electrode potential of -0.118 V. H₂ gas is bubbled at 298K and 1 atm pressure.

Solution: The electrode reaction is, $\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2} \text{H}_2$

$$E_{\text{H}^+/\text{H}_2} = E^0_{\text{H}^+/\text{H}_2} + \frac{RT}{nF} \ln[\text{H}^+]$$

$$-0.118 = 0 + 0.0591/1 \log [\text{H}^+]$$

$$-0.118 = 0 - 0.0591 \text{ pH}$$

$$\text{pH} = -0.118/-0.0591$$

$$\text{pH} = 2$$

- (iv) What is the potential of a half cell consisting of zinc electrode in 0.01M ZnSO₄ solution at 25°C. E° = 0.763V.

Solution: Concentration of Zn²⁺ ions = 0.01

$$[\text{Zn}^{2+}] = 0.01 \text{ M}$$

$$E_{\text{Zn}/\text{Zn}^{2+}} = E^0_{\text{Zn}/\text{Zn}^{2+}} + \frac{RT}{nF} \ln[\text{Zn}^{2+}]$$

$$E_{\text{Zn}/\text{Zn}^{2+}} = -0.763 + 0.0591/2 \log [0.01]$$

$$E_{\text{Zn}/\text{Zn}^{2+}} = -0.763 + 0.0591/2 (-2)$$

$$E_{\text{Zn}/\text{Zn}^{2+}} = -0.7902 - 0.0591$$

$$E_{\text{Zn}/\text{Zn}^{2+}} = -0.8221 \text{ V}$$

- (v) A Zn electrode containing 0.086 M ZnSO₄ was coupled with a copper electrode containing 0.069 M CuSO₄ to form a galvanic cell, (a) Write the cell representation, (b) Write the cell reactions, (c) Calculate the emf of the cell at 26.1° C,
Given, E°_{Zn/Zn²⁺} = -0.76 V and E°_{Cu/Cu²⁺} = 0.34 V,

Electrochemical Sensors – Potentiometric Sensors

Electrochemical sensors are devices that convert chemical information into an electrical signal. Among them, **potentiometric sensors** are widely used for selective and sensitive detection of ions and molecules.

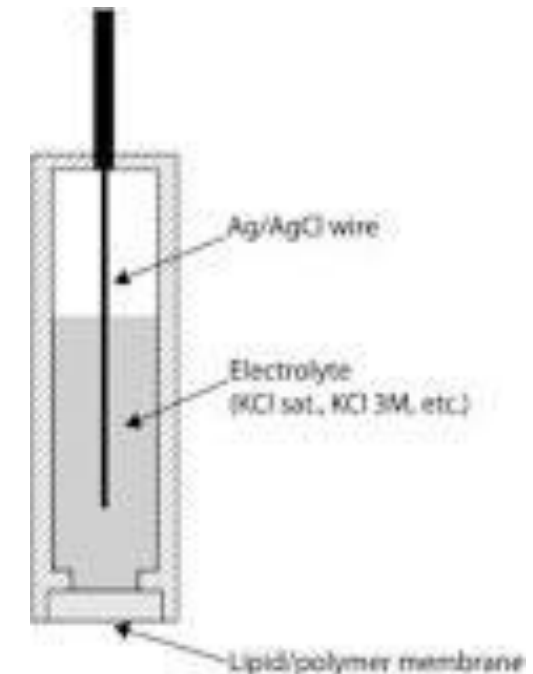
What are Potentiometric Sensors?

Potentiometric sensors measure the **electromotive force (EMF)** or potential difference between a **working electrode (indicator)** and a **reference electrode** without drawing any significant current. The measured potential is related to the concentration of a specific ion in the solution via the **Nernst equation**.

The typical example is pH meter. The potentiometric sensors have been developed on pathogen detection. Layer-by-layer technique is widely used for surface modifications

Key Features:

- High selectivity
- Minimal power requirement
- Useful for continuous and in situ measurements
- Output: voltage (mV), proportional to ion concentration



- Components:**
- 1.Indicator Electrode:** Selective to the ion of interest (e.g., glass electrode for H⁺).
 - 2.Reference Electrode:** Provides a stable reference potential (e.g., Ag/AgCl electrode).
 - 3.Ion-selective membrane:** Separates analyte and determines selectivity.

Sensor Type	Target Ion/Analyte	Indicator Electrode	Application
pH Sensor	H ⁺	Glass membrane electrode	Water quality, lab analysis, soil testing
Ion-Selective Electrode (ISE)	Na ⁺ , K ⁺ , Ca ²⁺ , NO ₃ ⁻ , F ⁻ etc.	Ion-selective membrane electrodes	Clinical diagnostics, food, agriculture
ISE for Fluoride	F ⁻	Lanthanum fluoride crystal electrode	Fluoride in drinking water
ISE for Nitrate	NO ₃ ⁻	PVC membrane with ionophore	Soil, water and environmental analysis
ISE for Potassium	K ⁺	Valinomycin-based membrane electrode	Blood electrolyte monitoring

Applications:

- Clinical diagnostics (blood electrolyte levels)
- Environmental monitoring (pH, heavy metal ions)
- Industrial process control
- Agricultural soil testing
- Food quality control

Electrochemical Sensors – Amperometric Sensors

Amperometric sensors are a type of electrochemical sensor that measure the **current** generated by the **redox reaction** of an analyte at a working electrode surface under a controlled potential. The current is **directly proportional** to the concentration of the analyte.

Key Features of Amperometric Sensors:

- Based on **Faradaic current** (oxidation/reduction)
- Requires an applied potential
- High sensitivity
- Rapid response time
- Used for electroactive species (those that undergo redox)

Working Principle:

When a constant potential is applied between the working and reference electrodes, the analyte undergoes **oxidation or reduction** at the working electrode, generating a current. This current is measured and is proportional to the concentration of the analyte.

$I \propto C$ Where:

- I = Measured current
- C = Concentration of analyte

Basic Components:

1.Working Electrode (WE): Where redox reaction of analyte occurs (e.g., Pt, Au, carbon).

2.Reference Electrode (RE): Maintains a constant potential (e.g., Ag/AgCl).

3.Counter Electrode (CE): Completes the circuit and allows current flow.

Sensor Type	Target Analyte	Electrode Material	Application
Glucose Sensor (biosensor)	Glucose	Platinum or carbon	Diabetes monitoring (blood glucose meters)
Oxygen Sensor (Clark Cell)	Dissolved O ₂	Platinum cathode	Medical ventilators, environmental testing
Hydrogen Peroxide Sensor	H ₂ O ₂	Carbon or modified electrodes	Food industry, medical diagnostics
Nitric Oxide Sensor	NO	Pt or Au	Biomedical research
Chlorine Sensor	Cl ₂ or hypochlorite	Gold or platinum	Water disinfection monitoring
Ethanol Sensor	Ethanol	Enzyme-modified electrode	Breath analyzers, fermentation monitoring

Example: Glucose Amperometric Biosensor

•**Enzyme:** Glucose oxidase (GOx)

•**Reaction:**



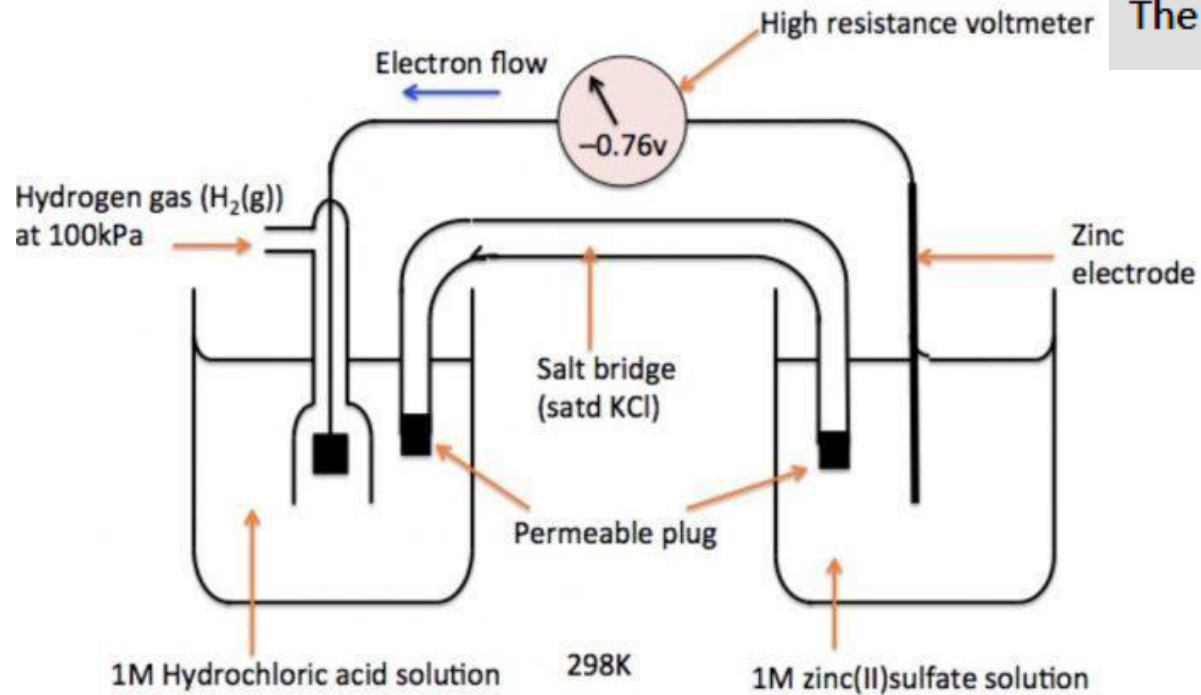
•**Detection:**

The H₂O₂ produced is oxidized at the electrode, generating a current.

Applications of Amperometric Sensors:

- Medical diagnostics (glucose, lactate, cholesterol)
- Environmental monitoring (oxygen, chlorine, pesticides)
- Industrial process control
- Food and beverage industry
- Biosensors (enzyme-based detection)

Determination of single electrode potential



The cell representation: $\text{Zn}/\text{Zn}^{2+}(1\text{M})//\text{HCl}(\text{H}^+, 1\text{M})/\text{H}_2(1\text{ atm})/\text{Pt}$

The cell potential was found to be 0.76 V. Now, we know that,

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}} = E_{\text{SHE}} - E_{\text{Zn}/\text{Zn}^{2+}}$$

$$0.76 = 0 - E_{\text{Zn}/\text{Zn}^{2+}}$$

$$E_{\text{Zn}/\text{Zn}^{2+}} = -0.76 \text{ V}$$

Electrochemical series

Definition: “Electrochemical series is an arrangement of various elements in the increasing order of their standard reduction potentials”.

Electrochemical Series

→

Thermodynamics measurement

STANDARD Reduction potential – H₂ as std

WEAK Oxidizing Agent	Oxidized sp ↔ Reduced sp	E°/V	STRONG Reducing Agent
TOP left • Low ↓ tendency gain e • Li⁺ + e → Li • E° Li = -3.04V • WEAK oxidizing Agent • Red NOT favourable (E° = -ve)	Li ⁺ + e- ↔ Li	-3.04	TOP right • High ↑ tendency lose e • Li → Li⁺ + e • E° Li = +3.04V • STRONG reducing Agent • Oxi favourable (E° = +ve)
	K ⁺ + e- ↔ K	-2.93	
	Ca ²⁺ + 2e- ↔ Ca	-2.87	
	Na ⁺ + e- ↔ Na	-2.71	
	Mg ²⁺ + 2e- ↔ Mg	-2.37	
	Al ³⁺ + 3e- ↔ Al	-1.66	
	Mn ²⁺ + 2e- ↔ Mn	-1.19	
	H ₂ O + e- ↔ H ₂ + OH ⁻	-0.83	
	Zn ²⁺ + 2e- ↔ Zn	-0.76	
	Fe ²⁺ + 2e- ↔ Fe	-0.45	
Ni ²⁺ + 2e- ↔ Ni	-0.26		
Sn ²⁺ + 2e- ↔ Sn	-0.14		
Pb ²⁺ + 2e- ↔ Pb	-0.13		
H ⁺ + e- ↔ H ₂	0.00		
Cu ²⁺ + e- ↔ Cu ⁺	+0.15		
SO ₄ ²⁻ + 4H ⁺ + 2e- ↔ H ₂ SO ₃ + H ₂ O	+0.17		
<u>Cu²⁺</u> + 2e- ↔ <u>Cu</u>	<u>+0.34</u>		
1/2O ₂ + H ₂ O + 2e- ↔ 2OH ⁻	+0.40		
Cu ⁺ + e- ↔ Cu	+0.52	BOTTOM left • High ↑ tendency gain e • F₂ + 2e → 2F⁻ • E° F₂ = +2.87V • STRONG oxidizing Agent • Red favourable (E° = +ve)	
1/2I ₂ + e- ↔ I ⁻	+0.54		
Fe ³⁺ + e- ↔ Fe ²⁺	+0.77		
Ag ⁺ + e- ↔ Ag	+0.80		
1/2Br ₂ + e- ↔ Br ⁻	+1.07		
1/2O ₂ + 2H ⁺ + 2e- ↔ H ₂ O	+1.23		
Cr ₂ O ₇ ²⁻ + 14H ⁺ + 6e- ↔ 2Cr ³⁺ + 7H ₂ O	+1.33		
1/2Cl ₂ + e- ↔ Cl ⁻	+1.36	WEAK Reducing Agent	
MnO ₄ ⁻ + 8H ⁺ + 5e- ↔ Mn ²⁺ + 4H ₂ O	+1.51		
Strong Oxidizing Agent	1/2F ₂ + e- ↔ F	+2.87	

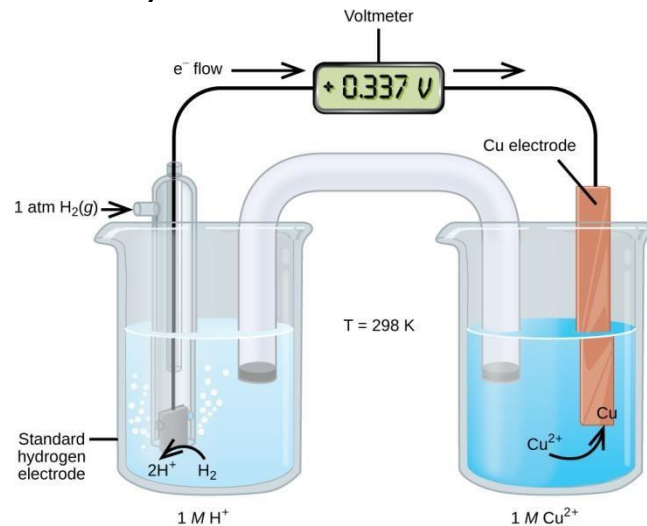
E°_{cell}/Cell Potential = EMF in volt
 EMF when half cell connect to SHE std condition
 Std potential written as std reduction potential

Electrochemical Conventions

1) Sign of the electrode potential

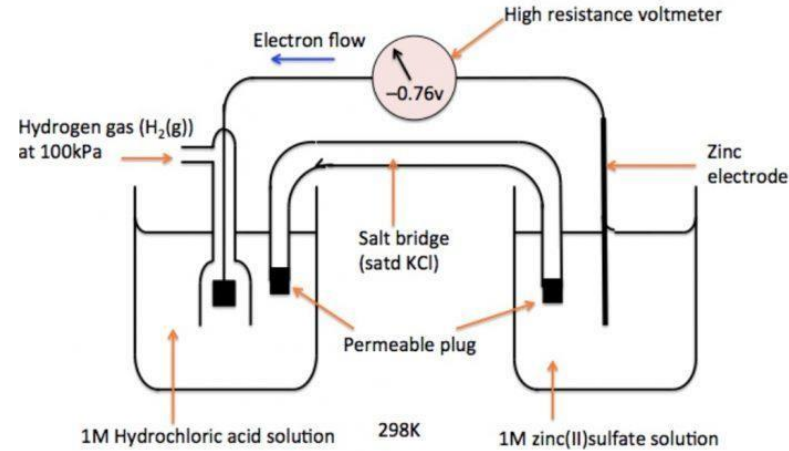
When combined with standard hydrogen electrode and

- i) If the electrode reaction Reduction, Then $E = +ve$ value is given
- ii) If the electrode reaction is Oxidation, then $E = -ve$ value is given



$$E_{\text{Cu}^{2+}/\text{Cu}} = + 0.34 \text{ V}$$

Reduction: Cu^{2+}

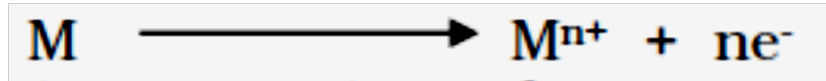


$$E_{\text{Zn}/\text{Zn}^{2+}} = - 0.76 \text{ V}$$

Oxidation: Zn

2) Representation of an Electrode

If an electrode acts as anode, i.e., if oxidation takes place at the electrode, then



The electrode is represented as $M_1/M_1^{n+} (C_1)$.

Example in Daniel cell, Zn electrode is represented as **Zn/Zn²⁺**

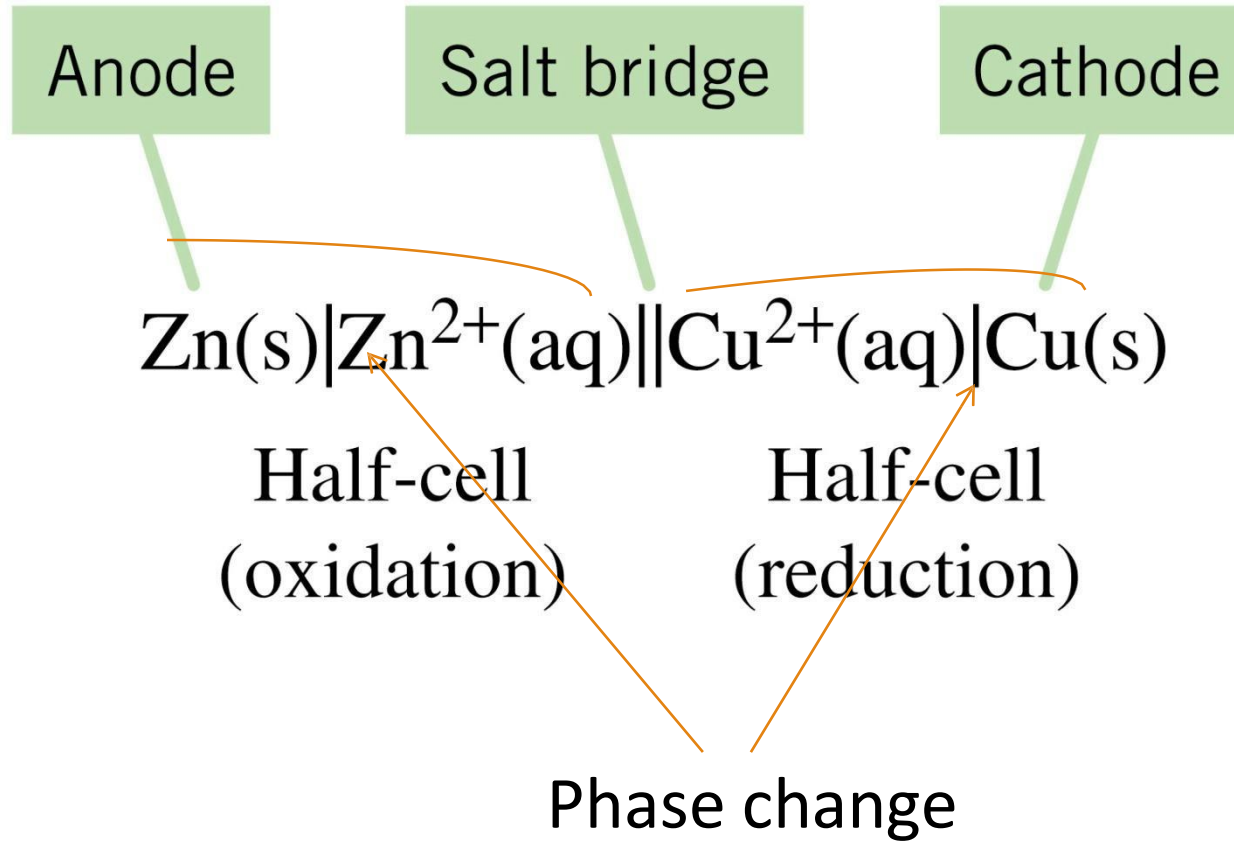
If an electrode acts as cathode, i.e., if Reduction takes place at the electrode, then



The electrode is represented as $M_2^{n+} (C_2) / M_2$

Example in Daniel cell, Cu electrode is represented as **Cu²⁺ /Cu**

3) Cell Representation



4) Writing Electrode reactions and cell reaction

The cell reaction is the sum of the electrode reactions as they occur in the cell. At the anode oxidation takes place and at the cathode reduction is the electrode reaction.

For example, in Daniel cell,

At anode: $\text{Zn(S)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$ (oxidation)

At cathode; $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(S)}$ (reduction)

Net cell reaction: $\text{Zn(S)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu(S)}$

5) Calculation of cell EMF

Emf or cell potential of a galvanic cell is given by, **$E_{\text{cell}} = E_{\text{c}} - E_{\text{a}}$ or $E_{\text{cell}} = E_{\text{R}} - E_{\text{L}}$**

For Daniel cell, $E_{\text{cell}} = E_{\text{R}} - E_{\text{L}}$

$$E_{\text{cell}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{Zn/Zn}^{2+}} = 0.34\text{V} - (-0.76)\text{V}$$

$E_{\text{cell}} = 1.1\text{V}$

Where $E_{\text{Cu}^{2+}/\text{Cu}}$ & $E_{\text{Zn/Zn}^{2+}}$ are standard electrode potentials (single electrode potentials) of copper and zinc half cells.

Types of Electrodes:

Metal- metal ion electrode

Metal dipped in its own salt ion solution Eg: Zn/Zn^{+2} , Cu/Cu^{2+}

Metal -metal salt ion electrode

Metal in contact with sparingly soluble salt of the same metal dipped in a solution containing anion of the salt

Eg: Calomel Electrode ($\text{Hg}/\text{HgCl}_2/\text{Cl}^-$), Silver -silver chloride electrode ($\text{Ag}/\text{AgCl}/\text{Cl}^-$)

Gas Electrode

Gas bubbling at electrode which is immersed in ionic solution in equilibrium with gas.

Eg: H_2 Electrode $\text{Pt}/\text{H}_2/\text{H}^+$.

Amalgam electrode

Metal amalgam dipped in its own metal ions Eg: Lead amalgam electrode ($\text{Pb-Hg}/\text{Pb}^{+2}$)

Oxidation-reduction electrode

Electrode potential arises from the presence of oxidized and reduced forms of the same substance in solution and the potential is picked by inert electrode like platinum. Eg: $\text{Pt}/\text{Fe}^{2+}, \text{Fe}^{3+}$

Ion selective electrodes:

Membrane electrodes in contact with a solution which it can exchange metal ions with solution Eg: Glass membrane electrode

Reference Electrodes

“An electrode with standard or constant electrode potential with which we can compare the potential of other electrodes is called a reference electrode”.

It can act both as an anode or cathode depending upon the nature of other electrode

The Reference Electrodes can be classified into two types

- i) Primary reference electrodes Ex: Standard hydrogen electrode
- ii) Secondary reference electrodes Ex: Calomel and Ag/AgCl electrodes

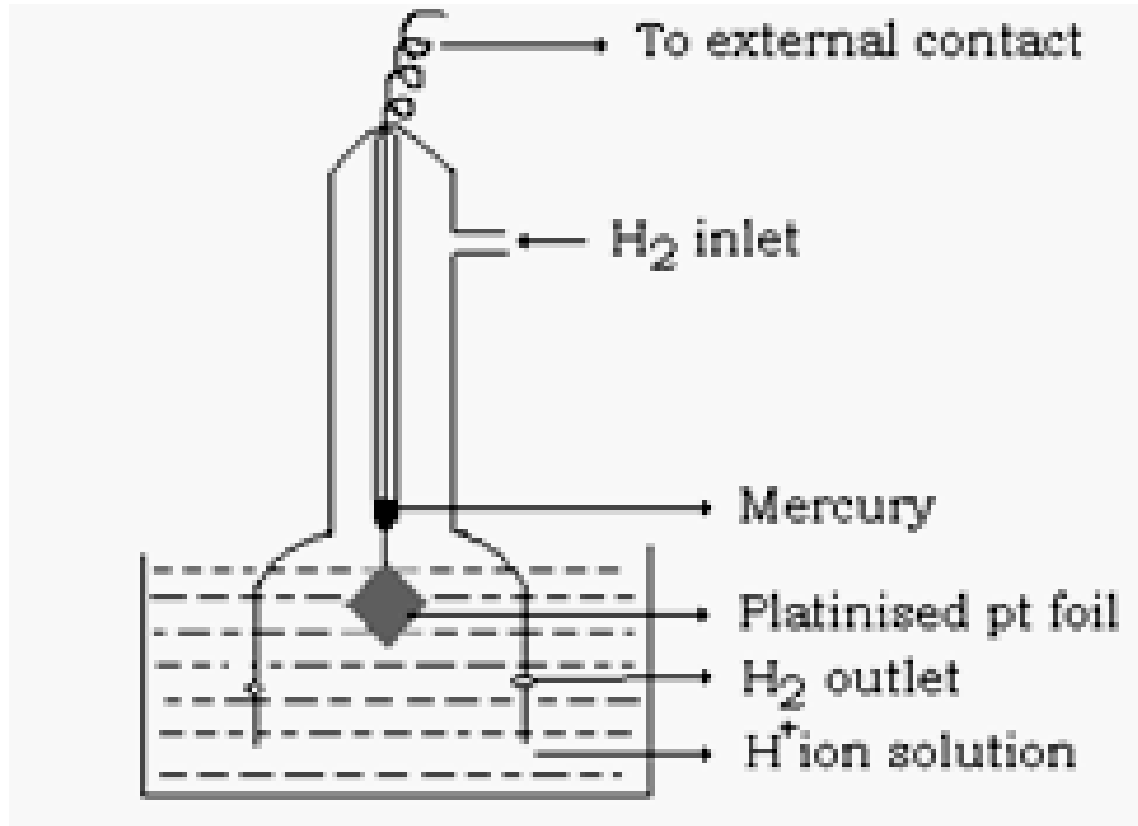
SHE has two main Limitations:

- i) construction of SHE is difficult and to maintain the concentration of H^+ as 1M and pressure H_2 gas at 1atm also difficult
- ii) Platinum electrode is poisoned by the impurities of the gas

Primary reference electrode:

- ✓ Standard Hydrogen Electrode (SHE) is primary reference electrode which is used for cell potential measurements.
- ✓ Its potential is taken as zero at all temperatures.

SHE Construction & working:



Electrode representation: $Pt/H_2 (g, 1 \text{ atm})/H^+ (1M)$

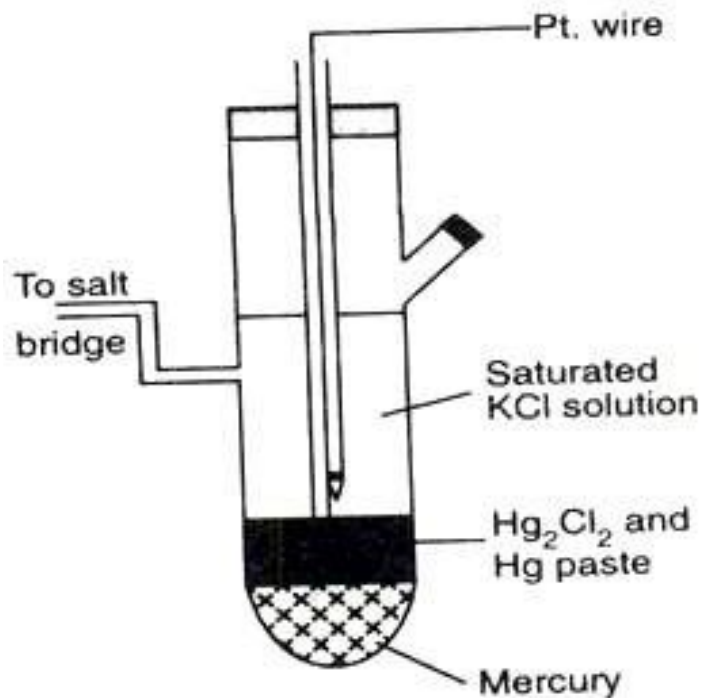
Reactions:

as anode: $H_2 (g, 1 \text{ atm}) \rightarrow 2H^+ (1M) + 2e^-$ ($E^0 = 0.00V$)

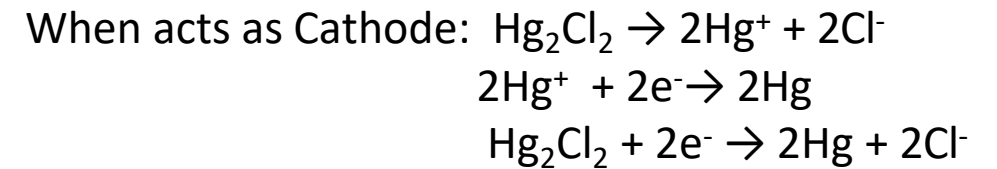
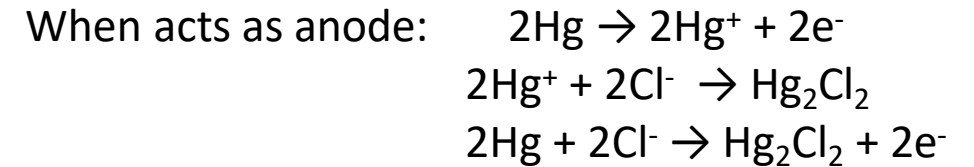
as cathode: $2H^+ (1M) + 2e^- \rightarrow H_2 (g, 1 \text{ atm})$ ($E^0 = 0.00V$)

Secondary reference electrode

Calomel electrode (Mercury-Mercurous chloride electrode):



Working and Reactions:



Calomel electrode is reversible with respect to Cl⁻ ion.

Electrode representation: $\text{Hg (l)}/\text{Hg}_2\text{Cl}_2(\text{s})/\text{KCl}(\text{sat})$

According to Nernst equation, Calomel electrode potential is given by,

$$E_{\text{Cal}} = E^{\circ}_{\text{Cal}} - \frac{RT}{2F} \ln [\text{Cl}^-]$$

$$E_{\text{Cal}} = E^{\circ}_{\text{Cal}} - 0.0591 \log [\text{Cl}^-] \text{ at } 298\text{K}$$

So the potential of Calomel electrode depends on concentration of the KCl solution.

Some of the most popularly used concentrations of KCl and corresponding single electrode (reduction) potentials on the hydrogen scale at 25°C are given below;

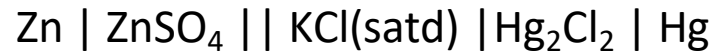
Concentration of KCl solution	Calomel electrode potential
0.1 N	0.3340 V
1.0N	0.2810 V
Sat.KCl	0.2422 V

Advantages of Calomel electrode:

- **simple to construct.**
- **Stable electrode & can be used over a long period.**
- **Cell potential or emf with calomel electrode is reproducible.**
- **Potential does not vary with temperature (It has least temperature coefficient)**

MEASUREMENT OF SINGLE ELECTRODE POTENTIAL USING CALOMEL ELECTRODE

To measure the electrode potential of zinc: To measure the potential of the Zn electrode, the Zn electrode is coupled with the SCE through a salt bridge. The anode and the cathode of the cell can be identified by connecting the electrodes to the appropriate terminals of the voltmeter. Proper measurements can be made only when the Zn electrode is connected to the –ve terminal and the calomel electrode to the +ve terminal of the voltmeter indicating that Zinc electrode is anode & the calomel electrode is a cathode.



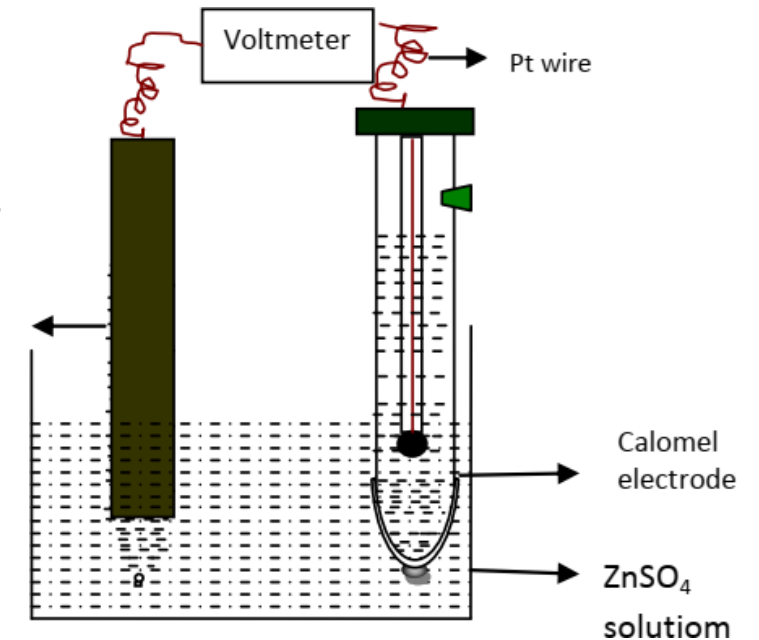
At anode: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$

At cathode: $\text{Hg}_2\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Hg} + 2\text{Cl}^-$

Overall reaction: $\text{Zn} + \text{Hg}_2\text{Cl}_2 \rightarrow \text{Zn}^{2+} + 2\text{Hg} + 2\text{Cl}^-$

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cathode}} - E_{\text{anode}} \\ &= E_{\text{SCE}} - E_{\text{Zn}^{2+}/\text{Zn}} \\ &= 0.2422\text{V} - E_{\text{Zn}^{2+}/\text{Zn}}^0 \\ E_{\text{Zn}^{2+}/\text{Zn}}^0 &= 0.2422 - E_{\text{cell}} \\ E_{\text{Zn}^{2+}/\text{Zn}}^0 &= 0.2422 - 1.001 \end{aligned}$$

$$E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$$



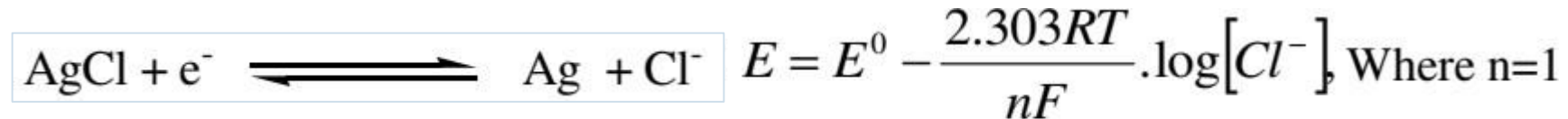
Silver- Silver Chloride electrode

- Silver-Silver chloride is also a metal-metal salt ion electrode.
- Silver and its sparingly soluble salt silver chlorides are in contact with a solution of chloride solution ions.
- Generally a silver wire is coated with AgCl and dipped in a solution of KCl .

Cell representation is as follows

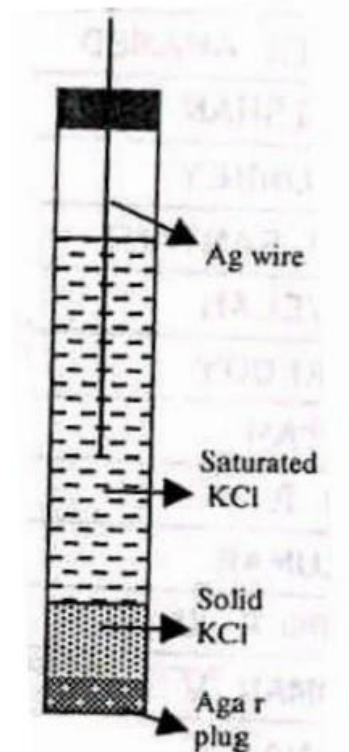


Net half cell reaction is



$$E = E^0 - \frac{2.303RT}{F} \log[\text{Cl}^-]$$

$$E = E^0 - 0.0591 \log[\text{Cl}^-] \text{ at } 298\text{K}$$



Therefore electrode potential of calomel electrode is depending upon the concentration of KCl.

- ✓ electrode is reversible with chloride ions.**
- ✓ potential of the calomel electrode depends on the concentration of the KCl.**

For 1N solution, the electrode potential is 0.223V and for saturated solution is 0.199V at 298K

Applications:

- 1. Used as secondary reference electrode in ion selective electrode.**
- 2. In determining the distribution of potential on the ship hull and pipe lines.**

BATTERY

- *A battery is an Electrochemical cell or often several electrochemical cells which are connected in series to produce a constant voltage to run portable goods*
- *It is a compact device consisting of two or more galvanic cells connected in series or parallel or both.*
- *It stores chemical energy in the form of active materials and on demand converts it into electrical energy through redox reactions*

Ex. Batteries are used in calculators, digital watches, pace makers, hearing aids, portable computers, electronically controlled cameras, digital watches, stand by power supplies, emergency lighting and electroplating, telecommunication, military & space applications.

PRINCIPAL COMPONENTS OF A BATTERY:

1. An **anode** where oxidation
2. A **cathode** where reduction occurs
3. An **electrolyte** , which is ironically conducting
4. A **separator** to separate anode and cathode compartments

Types: 3

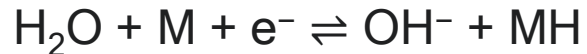
- 1) Primary Battery is the one in which cell reactions are irreversible. (Single use)
E.g. Leclanche cell or Zn-Carbon Battery or Dry cell .
- 2) Secondary Battery is the one in which cell reactions can be reversed by passing external emf in opposite direction. i.e. it can be used for many cycles of charging and discharging.
E.g. Lead – Acid battery, Ni-Cd etc.
- 3) Flow battery is the one in which all the constituents of the battery flow through out the battery
E.g. Hydrogen-Oxygen Fuel cell.

Metal hydrides

A **nickel–metal hydride battery (NiMH or Ni–MH)** is a type of [rechargeable battery](#). The chemical reaction at the positive electrode is similar to that of the [nickel-cadmium cell](#) (NiCd), with both using [nickel oxide hydroxide](#) (NiOOH). However, the negative electrodes use a hydrogen-absorbing [alloy](#) instead of [cadmium](#). NiMH batteries can have two to three times the capacity of NiCd batteries of the same size, with significantly higher [energy density](#), although only about half that of [lithium-ion batteries](#).^[6]

They are typically used as a substitute for similarly shaped non-rechargeable [alkaline batteries](#), as they feature a slightly lower but generally compatible cell voltage and are less prone to [leaking](#).^{[7][8]}

The negative electrode reaction occurring in a NiMH cell is



On the positive electrode, nickel oxyhydroxide, NiO(OH), is formed:



The reactions proceed left to right during charge and the opposite during discharge. The metal M in the negative electrode of a NiMH cell is an [intermetallic](#) compound.

^[19]

NiMH cells have an [alkaline electrolyte](#), usually [potassium hydroxide](#). The positive electrode is nickel hydroxide, and the negative electrode is hydrogen in the form of an interstitial metal hydride.^[20] Hydrophilic [polyolefin](#) nonwovens are used for separation.^[2]

Nickel-Metal Hydride (NiMH) batteries are **rechargeable secondary cells**. They are widely used in portable electronics, hybrid vehicles, and button cell applications. NiMH is an improvement over nickel-cadmium (NiCd) batteries, offering **higher capacity** and **less environmental toxicity**.

2. Working Principle of NiMH Battery

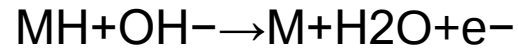
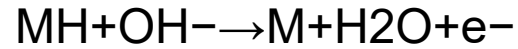
NiMH batteries operate based on **redox reactions** involving:

- A **metal hydride (MH)** alloy as the **negative electrode (anode)**.
- Nickel oxyhydroxide (NiOOH)** as the **positive electrode (cathode)**.
- An **alkaline electrolyte** (usually potassium hydroxide, KOH).

3. Cell Reactions

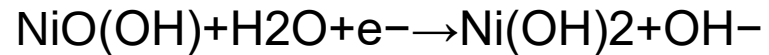
At the Anode (Negative Electrode):

Metal hydride gets oxidized:

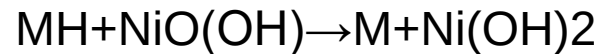


At the Cathode (Positive Electrode):

Nickel oxyhydroxide gets reduced:



Overall Cell Reaction:



This reaction is **reversible**. During charging, the reverse reaction occurs.

4. Features of NiMH Batteries

- **Voltage per cell:** ~1.2 V
- **Energy density:** Higher than NiCd
- **Cycle life:** 500–1000 charge cycles
- **No memory effect** (less pronounced than NiCd)
- **Environmentally safer** (no cadmium)

5. Button Cells (NiMH Type)

NiMH technology is also adapted to **miniature button cells**, typically used in:

- Hearing aids
- Watches
- Medical devices
- Calculators

Advantages of NiMH Button Cells:

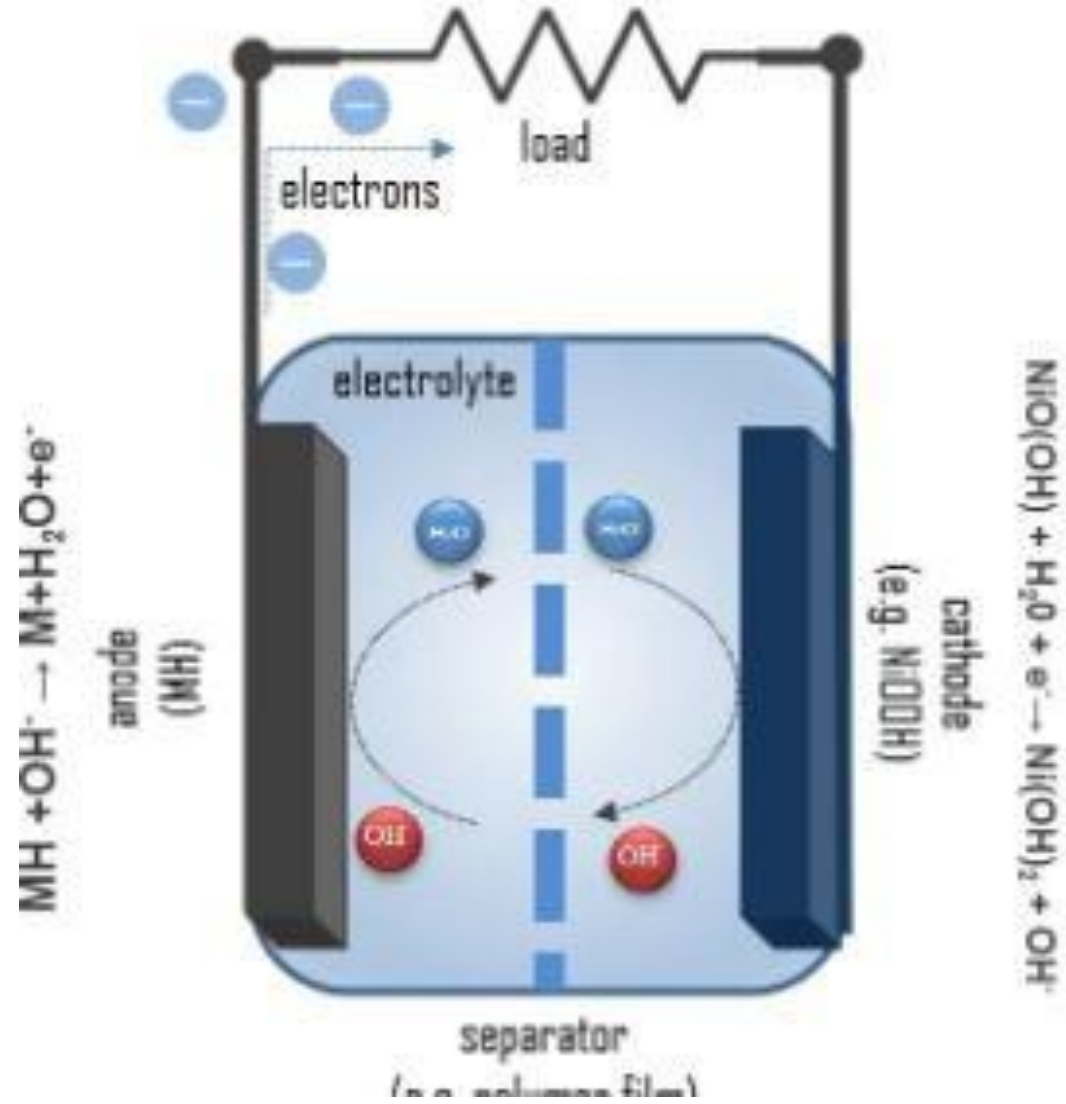
- Rechargeable and compact
- More environmentally friendly than silver oxide or lithium button cells
- Suitable for low-to-moderate power applications

Component	Material Used
Anode (–)	Metal Hydride Alloy (e.g., LaNi ₅ H ₆)
Cathode (+)	Nickel Oxyhydroxide (NiO(OH))
Electrolyte	Potassium Hydroxide (KOH)
Cell Voltage	~1.2 V

Advantages and Considerations:

- NiMH batteries offer higher energy density than nickel-cadmium (NiCd) batteries, while also being environmentally friendlier due to the absence of cadmium.
- Their performance can be significantly affected by temperature, with discharge capacity decreasing at very low or high temperatures.
- While once a replacement for NiCd batteries, NiMH technology is now gradually being replaced by lithium-ion batteries in many applications due to lithium-ion's higher energy density.

NiMH Battery



Primary Cells – Alkali Metal Sulphide Batteries

Overview:

Primary cells are electrochemical cells that are designed for single-use, as the chemical reactions that occur within them are not easily reversible. Among various types, **alkali metal sulphide batteries** are a special category that uses alkali metals (such as lithium or sodium) as the anode and a sulphide-based compound as the cathode.

Key Examples:

1.Lithium/Iron Disulfide (Li/FeS₂) – commonly used in **Energizer Ultimate Lithium batteries** (AA/AAA sizes).

2.Sodium/Sulphur (Na/S) – mostly used in **high-temperature rechargeable cells**, but modified forms can act as primary cells.

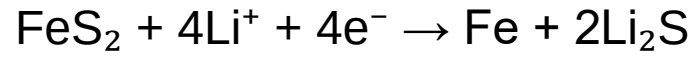
The **lithium–sulfur battery** (Li–S battery) is a type of [rechargeable battery](#). It is notable for its high [specific energy](#).^[2] The low [atomic weight](#) of [lithium](#) and moderate atomic weight of [sulfur](#) means that Li–S batteries are relatively light (about the density of water). They were used on the longest and highest-altitude unmanned [solar-powered](#) aeroplane flight (at the time)

Chemical Reaction Example – Li/FeS₂ Cell:

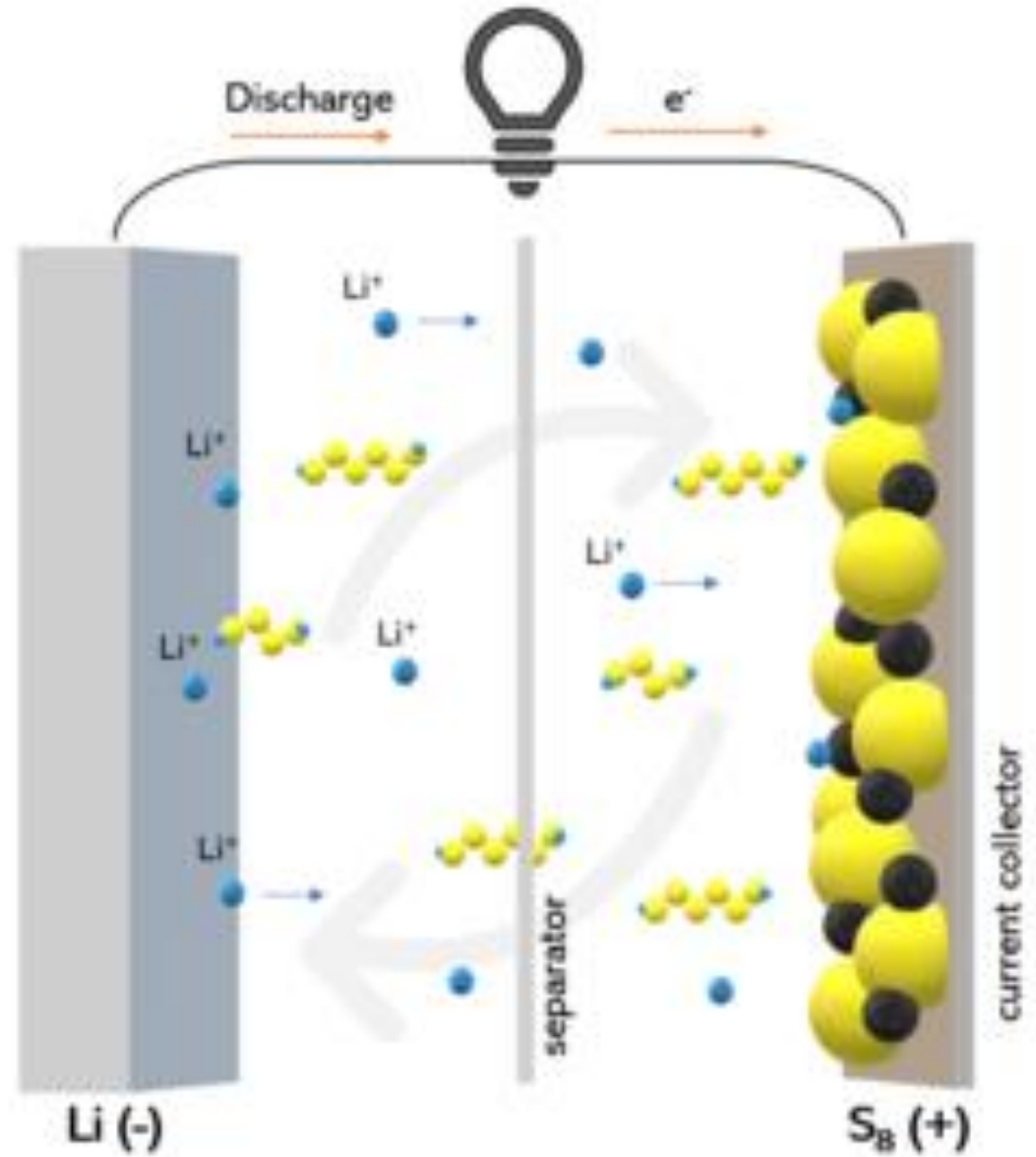
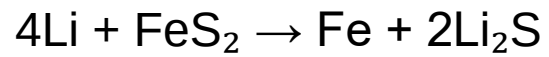
Anode reaction:



Cathode reaction:



Overall cell reaction:



Advantage of primary cell

- High Theoretical Capacity:**

- Sulfur offers a high theoretical capacity, making these batteries attractive for high-energy applications.

- Abundance and Cost:**

- Alkali metals like sodium and potassium are more abundant and cost-effective than lithium, making Na-S and K-S batteries particularly interesting for future energy storage.

- Challenges:**

- The main challenge in alkali metal sulfide batteries is the "shuttle effect," where soluble polysulfides formed during discharge cycle dissolve in the electrolyte and shuttle between electrodes, reducing battery performance and lifespan.

Fuel Cell

A ***fuel cell*** is a device that converts the chemical energy of a fuel (hydrogen, natural gas, methanol, gasoline, etc.) and an oxidant (air or oxygen) into electricity.

- Anode
- Cathode
- Electrolyte

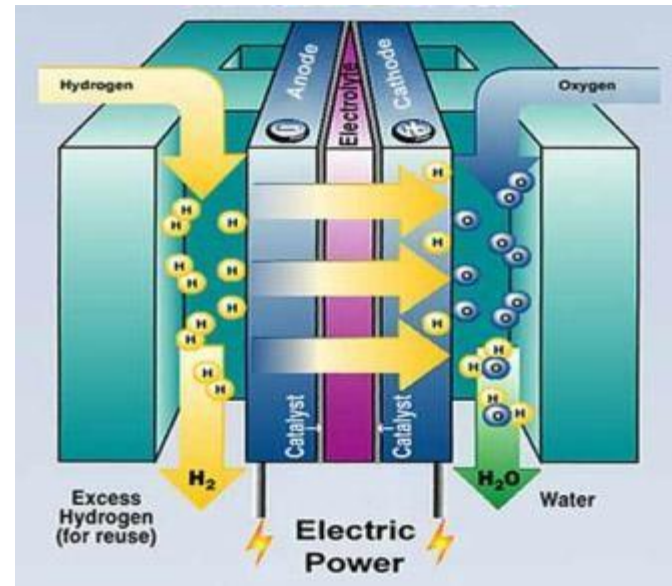
Fuel | electrode | electrolyte | electrode | oxidant

Anode:

Fuel $\rightarrow$ oxidized product + ne

Cathode:

oxidant + $ne \rightarrow$ reduced product



$$\Delta G = -nFE$$

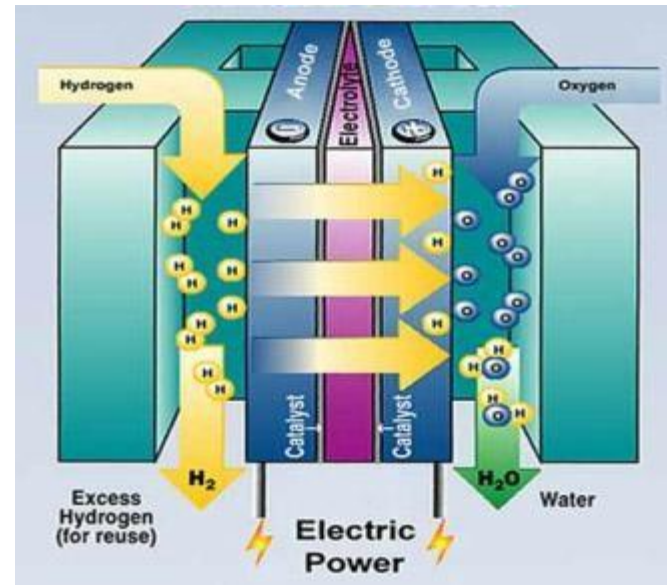
Designing Fuel Cell

Fuel | electrode | electrolyte | electrode | oxidant

At the anode, a catalyst oxidizes the fuel (e.g., H_2), turning the fuel into a cation and an electron.

Electrolyte: Ions can pass through it, but not electron. The electrons travel through a wire creating the electric current.

Cathode: The ions travel through the electrolyte to the cathode. Once reaching the cathode, the ions are reunited with the electrons and the two react with a third chemical (e.g., O_2) to create water or carbon dioxide.



Fuel Cell

Merits of Fuel Cell

1. High efficiency for conversion of chemical to electrical energy
2. Recharging of the fuel cell is not required
3. Less attention is needed
4. Harmless waste products and eco-friendly

Classification of Fuel Cells

1. Low temperature fuel cells ($< 100\text{ }^{\circ}\text{C}$)
2. Moderate temperature fuel cells (100-250 $^{\circ}\text{C}$)
3. High temperature fuel cells ($< 500\text{ }^{\circ}\text{C}$)

Types of Fuel Cells

1. Alkaline fuel cells (AFC)
2. Phosphoric acid fuel cells (PAFC)
3. Molten carbonate fuel cells (MCFC)
4. Polymer electrolyte membrane fuel cells (PEMFC)
5. Solid oxide fuel cells (SOFC)
6. Biochemical fuel cells (BCFC)

<u>Type</u>	<u>ions</u>	<u>Electrolyte</u>
PEM	H+	solid polymer (e.g. Nafion)
PAFC	H+	phosphoric acid (H ₃ PO ₄)
Alkaline	OH-	potassium hydroxide (KOH) in water
Solid Oxide	O ²⁻	yttrium-stabilized zirconia (YSZ)
Molten Carbonate	CO ₃ ²⁻	lithium and potassium carbonate mixture (Li ₂ CO ₃ and K ₂ CO ₃)

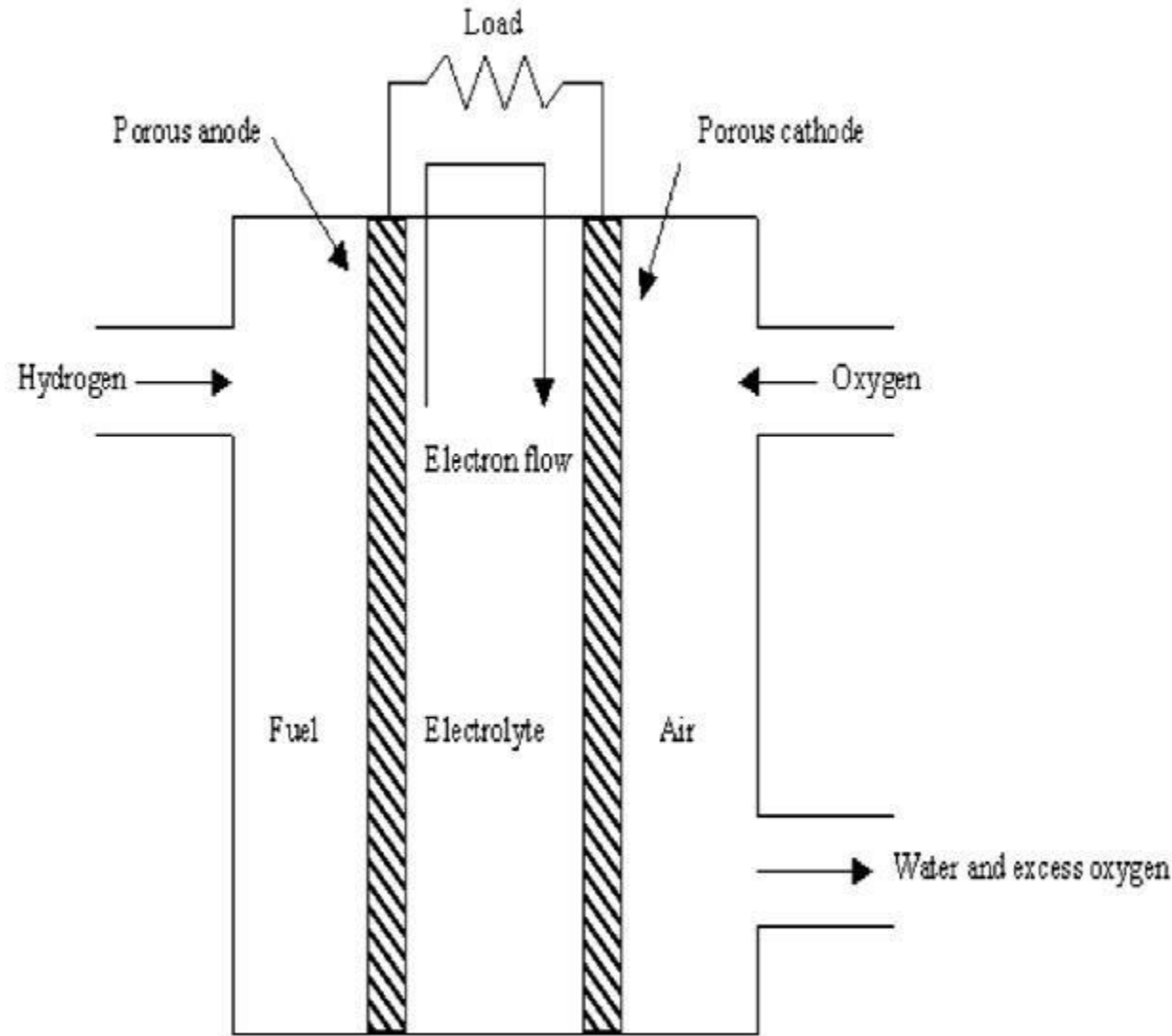
Chemical Reactions in Fuel cells

Fuel Cell	Anode Reaction	Cathode Reaction	Overall Reaction
Proton Exchange	$2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{l})$
Alkaline	1) $2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$ 2) $4\text{H}^+ + 4(\text{OH})^- \rightarrow 4\text{H}_2\text{O}$	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4(\text{OH})^-$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{l/g})$
Phosphoric Acid	$2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{g})$
Molten Carbonate	1) $2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$ 2) $4\text{H}^+ + 2\text{CO}_3^{2-} \rightarrow 2\text{H}_2\text{O} + 2\text{CO}_2$	$\text{O}_2 + 2\text{CO}_2 + 4\text{e}^- \rightarrow 2\text{CO}_3^{2-}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{g})$
	1) $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$ (reformation) 2) $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$ (water-gas shift) 3) $4\text{H}_2 + \text{CO}_2 \rightarrow \text{CO}_2 + 8\text{H}^+ + 8\text{e}^-$ 4) $\text{CO}_2 + 8\text{H}^+ + 4\text{CO}_3^{2-} \rightarrow 5\text{CO}_2 + 4\text{H}_2\text{O}$	$2\text{O}_2 + 4\text{CO}_2 + 8\text{e}^- \rightarrow 4\text{CO}_3^{2-}$	$\text{CH}_4 + 2\text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$
Solid Oxide	1) $2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$ 2) $4\text{H}^+ + 2\text{O}^{2-} \rightarrow 2\text{H}_2\text{O}$	$\text{O}_2 + 4\text{e}^- \rightarrow 2\text{O}^{2-}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{g})$
	1) $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$ (reformation) 2) $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$ (water-gas shift) 3) $4\text{H}_2 + \text{CO}_2 \rightarrow \text{CO}_2 + 8\text{H}^+ + 8\text{e}^-$ 4) $\text{CO}_2 + 8\text{H}^+ + 4\text{O}^{2-} \rightarrow \text{CO}_2 + 4\text{H}_2\text{O}$	$2\text{O}_2 + 8\text{e}^- \rightarrow 4\text{O}^{2-}$	$\text{CH}_4 + 2\text{O}_2 \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$

*Proton-exchange membrane fuel cells (PEMFC), also known as polymer electrolyte membrane (PEM) fuel cells

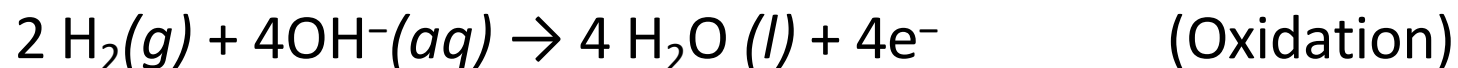
Construction:

- ✓ Galvanic cell where there is a continuous supply of reactants.
- ✓ Consists of two inert porous electrodes made of graphite impregnated with finely divided Pt and 25% KOH solution as electrolyte.
- ✓ H_2 gas is bubbled through one inert electrode, acts as anode.
- ✓ O_2 gas is bubbled through another electrode, acts as cathode.
- ✓ The hydrogen-oxygen fuel cell produces water as a product and hence is an ideal power source for zero-emission vehicles. Hence it is called an eco friendly battery.

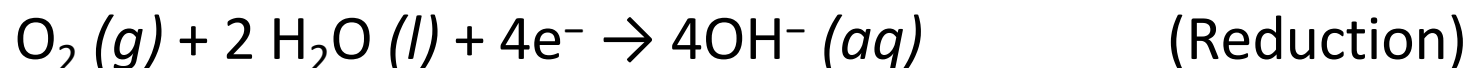


Working: Redox reaction in a Hydrogen- oxygen fuel Cell

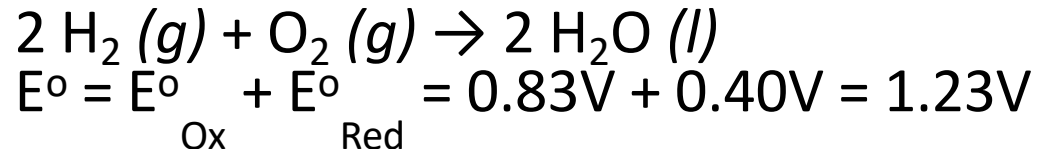
Anode:



Cathode:



Net Reaction:



Uses:

Auxiliary energy source in space vehicles, submarines and other military vehicles

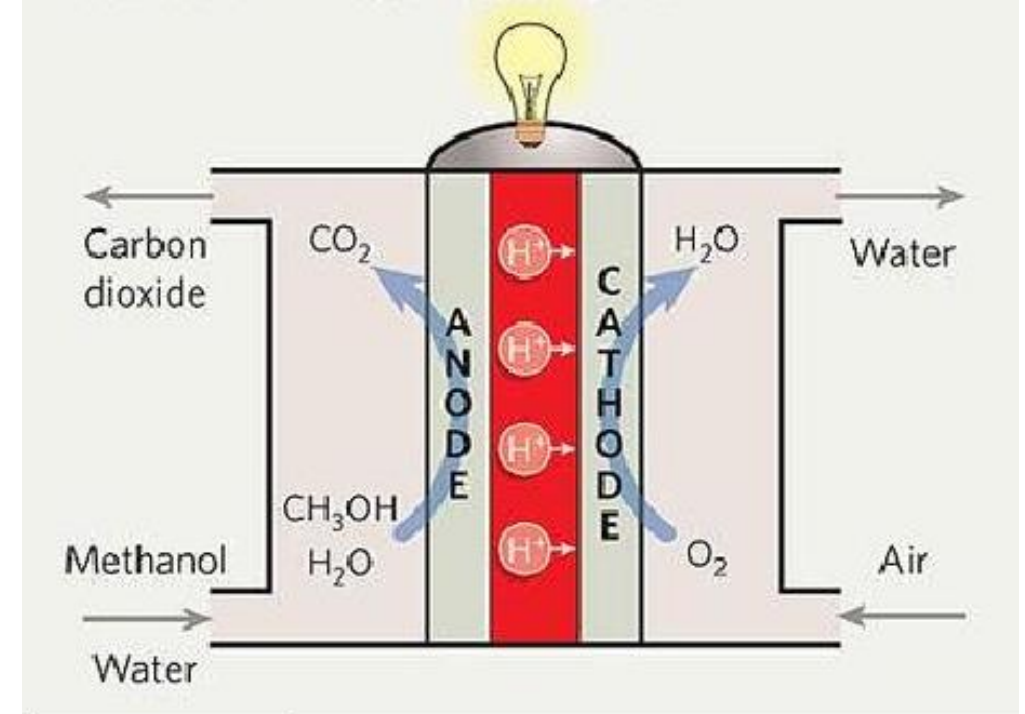
METHYL ALCOHOL – OXYGEN FUEL CELL

Construction:

- It consists of two electrodes made up of Pt in between the electrodes H_2SO_4 is placed as an electrolyte.
- Methanol and H_2SO_4 is supplied at the anode and pure O_2 gas is supplied at the cathode.
- Methanol is oxidized to CO_2 & H_2O with the liberation of electrical energy.
- Cell delivers an EMF of 1.2 V. The cell reactions are as follows

Uses:

Used in Military applications and also for large scale power production stations.



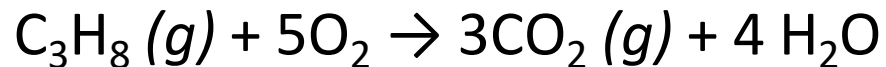
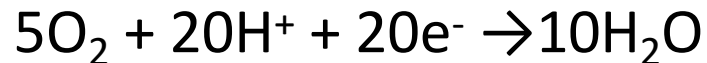
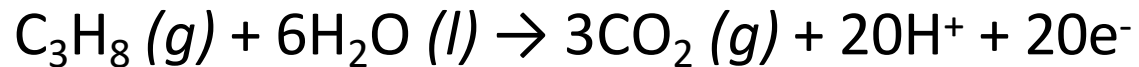
	Equation
Anode	$\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow 6 \text{H}^+ + 6 \text{e}^- + \text{CO}_2$ oxidation
Cathode	$\frac{3}{2} \text{O}_2 + 6 \text{H}^+ + 6 \text{e}^- \rightarrow 3 \text{H}_2\text{O}$ reduction
Overall reaction	$\text{CH}_3\text{OH} + \frac{3}{2} \text{O}_2 \rightarrow 2 \text{H}_2\text{O} + \text{CO}_2$ redox reaction

PROPANE– OXYGEN FUEL CELL

Construction:

- It consists of 2 electrodes made up of Pt in between electrodes H_2SO_4 is placed as a electrolyte.
- Propane and H_2SO_4 is supplied at the anode and pure O_2 gas is supplied at the cathode.
- Propane is oxidized to CO_2 & H_2O with the liberation of electrical energy.
- Cell delivers an emf of 1.20 V. The cell reactions are as follows.

Working:



Uses:

Used in Military applications and also for large scale power production stations.